Greenfield or Brownfield?
FDI Entry Mode and Intangible Capital*

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Abstract

When a firm invests abroad, it can either establish a new facility through greenfield foreign direct investment (GFDI) or purchase a local firm through cross-border mergers and acquisitions (M&A). This paper studies how intangible capital shapes this entry-mode choice and its consequences for host countries. I develop a general equilibrium search model in which foreign projects choose between M&A and GFDI. The model predicts that projects with higher levels of intangible capital are more likely to enter through GFDI. I test this prediction using a novel US project-level dataset and find empirical support. Consistent with the model, M&A is also more likely in host-country–industry markets with thicker pools of local firms and richer local intangible assets. I then use the calibrated model to assess whether the decentralized FDI composition is efficient for the host country. Advanced economies lie close to the efficient margin, whereas non-advanced economies receive too little M&A because search frictions are high. A balanced-budget policy that subsidizes completed M&A raises host welfare, but the gains are limited by target scarcity.

Keywords: FDI, Intangible capital, Cross-border M&A, Greenfield FDI

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1 Introduction

Multinationals and their foreign affiliates generate one-third of global GDP and account for two-thirds of international trade.[1] In light of their economic importance, many governments have offered subsidies and tax incentives to attract foreign direct investment (FDI). Host countries receive FDI in two main modes: *greenfield foreign direct investment* (GFDI), in which new facilities are developed by foreign firms, and *brownfield investment*, also called *cross-border mergers and acquisitions* (M&A), in which local firms are purchased by foreign firms. Both modes of investment are economically important. In recent years, there have been more than twice as many GFDI projects as M&A deals, whereas the total value of these transactions has been virtually the same (UNCTAD, 2019).

Policymakers tend to assume that new facilities create more jobs than acquisitions of existing facilities, and therefore almost all investment promotion agencies encourage GFDI rather than M&A (Caves, 1996).[2] While policy attention has largely focused on GFDI, a substantial literature shows that acquired firms often experience productivity improvements (Schoar, 2002; Arnold and Javorcik, 2009; Li, 2013; Dimopoulos and Sacchetto, 2017) and foreign acquisitions can also foster innovation and the adoption of new organizational practices at acquired firms (Bloom et al., 2012; Guadalupe et al., 2012). In this context, it is of first-order policy importance to understand how foreign firms decide whether to pursue GFDI or M&A. However, empirically we have little understanding of the factors that determine FDI entry mode, partly due to a lack of appropriate data. Theoretically, existing models are not tailored to analyze firms' FDI mode choices and their welfare consequences for host countries. I address this gap by asking two questions: (1) how do firms choose between GFDI and M&A, and (2) how does this choice affect the host economy?

I organize the analysis around a single premise: the key difference between GFDI and M&A is the role of intangible capital, such as brands, intellectual property, customer relationships, and supplier networks. One of the defining characteristics of intangible capital is its non-rivalry in use (Crouzet et al., 2022). That is, unlike physical capital, intangible capital can be used in multiple locations simultaneously. Because of this characteristic, intangible capital plays an important role in FDI (Markusen, 1995; Burstein and Monge-

[1] This information comes from the OECD analytical AMNE database in 2016. I refer to the VOX EU CEPR column, "Multinational enterprises in the global economy: Heavily discussed, hardly measured," published on September 25, 2019.

[2] For example, in the survey of OECD and Latin America and the Caribbean countries, around 80% of agencies target GFDI, while around 5% target M&A (Volpe Martincus and Sztajerowska, 2019). In the survey of OECD countries, no agency targets M&A (OECD, 2018).

Naranjo, 2009; McGrattan and Prescott, 2009; McGrattan and Prescott, 2010). If investing firms make intensive use of their own intangible capital, they are also likely to use those intangibles in foreign markets, thus relying more on GFDI and less on M&A.[3]

Motivated by this premise, I develop a general equilibrium search model of a small open host economy in which heterogeneous foreign projects choose between GFDI and M&A. Foreign projects share a common productivity component but differ in intangible capital; both components are transferable across countries. The complementarity between productivity and intangible capital generates a trade-off in the M&A decision. Under M&A, a foreign project acquires a local target's intangibles and upgrades them with its superior productivity, but integrates only a fraction of its own intangibles, with the realized integration quality drawn after matching. Acquisitions also require costly search and matching in a frictional merger market, so a foreign project that prefers M&A may still fail to find a partner. The model delivers two testable implications. First, conditional on investing abroad, projects with higher intangible capital are more likely to enter through GFDI, because they have less to gain from a target's intangibles and more to lose from imperfect integration. Second, because M&A requires a successful match, the relative attractiveness of M&A rises with the quality of the local matching environment. For example, a higher matching probability or a lower search cost shifts projects toward M&A.

I test these implications using a novel US project-level dataset that links GFDI projects (fDi Markets), the universe of cross-border M&A deals (SDC Platinum), and financial information on US publicly listed firms (Compustat). I measure each firm's internally generated intangible capital following Peters and Taylor (2017) and Ewens et al. (2025). The data support both model implications. Projects whose parent firms have lower levels of intangible capital are more likely to enter through M&A. The results are robust to alternative methods for constructing intangible capital. Turning to the host-country environment, M&A is more likely in destination-industry markets with thicker pools of local firms and richer local intangible assets. It is also less likely in markets with greater geographic and linguistic barriers, consistent with higher search and information frictions.

I then use the calibrated model to study host-country welfare. The decentralized equilibrium need not be efficient: foreign projects choose search and entry mode to maximize their own repatriated payoffs, whereas the host country values the resulting composition of foreign and local varieties through household income and the price index. Comparing the

[3]For example, multinational firms such as Walmart and Kellogg with established global brands—a type of intangible capital—will likely pursue GFDI. Firms that do not have well-known brands or reputations would seek instead to acquire local brands (DePamphilis, 2019).

two yields an inefficiency wedge whose sign determines whether the host receives too much or too little M&A. Calibrating the model separately to advanced and non-advanced destinations, I find that advanced economies lie close to the efficient margin, whereas non-advanced economies receive too little M&A primarily because search costs in the merger market are high. Motivated by this inefficiency, I study a balanced-budget policy in the non-advanced economy that subsidizes each completed M&A and finances it with a tax on GFDI. The welfare gain is positive but small, and the reason is target scarcity: only a thin subset of local producers is effectively acquirable, so additional search is absorbed mainly by a lower matching probability rather than by completed mergers. The lesson is therefore not simply that host countries should subsidize M&A, but that the effectiveness of transaction subsidies depends on the availability of acquirable targets.

This paper develops a new model of FDI mode choice, focusing on how intangible capital affects firms' choice between GFDI and cross-border M&A. My work is primarily related to the literature on foreign market entry, including Helpman et al. (2004), Ramondo and Rodríguez-Clare (2013), and Tintelnot (2017). Unlike these studies, which consider firms' exporting and FDI decisions, my research focuses on the mode of FDI—that is, whether firms enter foreign markets through GFDI or cross-border M&A.

Compared with the extensive literature on firms' exporting and FDI decisions, fewer studies examine FDI mode choice. The studies most relevant to my research are Nocke and Yeaple (2007; 2008).[4] In particular, Nocke and Yeaple (2008) show that firms do not perceive the two modes of FDI as perfect substitutes and propose an assignment theory of multinationals' FDI mode choice. My paper differs from their study in three ways. First, I provide a comprehensive empirical analysis using a larger and more up-to-date dataset than Nocke and Yeaple (2008). Using this new project-level dataset, I provide the first empirical evidence that intangible assets are strongly correlated with firms' FDI mode choices. The results are robust to the additional controls used by Nocke and Yeaple (2008), namely sales and value added per worker. Second, I construct an equilibrium search model of mergers that is consistent with empirical features from the data. I characterize search decisions based on the Diamond-Mortensen-Pissarides (DMP) model (Pissarides, 2000). I follow Rhodes-Kropf and Robinson (2008) and David (2021), who study domestic M&A activity, to incorporate

[4]Other studies on FDI mode choice focus on geographical and cultural barriers (Davies et al., 2018); matching of core competencies between acquirer and target firms (Díez and Spearot, 2014); vertical and horizontal FDI (Ramondo, 2016); total factor productivity in developed and developing countries (Ashraf et al., 2016); and two greenfield ownership choices, wholly owned affiliates or joint ventures (Raff et al., 2012). Similarly, my research builds on a theoretical literature that aims to predict how FDI mode choice affects welfare (Norbäck and Persson, 2007; Kim, 2009; Bertrand et al., 2012).

search frictions into my model of the international merger market. Third, I analyze the welfare implications of equilibrium FDI patterns. In contrast to Nocke and Yeaple (2008), who consider a frictionless perfectly competitive merger market, I study the role of frictions that have been emphasized in the M&A literature. My model allows me to characterize the host-country inefficiency in the laissez-faire equilibrium.

Finally, this paper underlines the critical role that FDI plays in transferring firms' technology and knowledge (i.e., their intangible capital) across countries, a fact well-documented in the literature (Teece, 1977; Dunning, 1980; Burstein and Monge-Naranjo, 2009; McGrattan and Prescott, 2010; Bloom et al., 2012; Keller and Yeaple, 2013; Arkolakis et al., 2018; Bilir and Morales, 2020).[5] I connect firms' knowledge transfer to FDI mode choices and show that M&A can raise host welfare in non-advanced economies, although the gains are limited by the availability of acquirable targets.

The remainder of the paper is organized as follows. Section 2 presents the model of FDI entry-mode choice. Section 3 introduces the data, and Section 4 tests the model's implications. Section 5 calibrates the model and presents the welfare and counterfactual policy analysis. Section 6 concludes.

2 A Model of FDI Entry Mode Choice

This section introduces a model of FDI entry mode choice. The model provides a framework to analyze how heterogeneity in intangible capital shapes the decision to invest abroad and the choice between greenfield investment (GFDI) and cross-border M&A. The framework is later used to study welfare and policy implications.

The model integrates three strands of the literature. First, in the spirit of Helpman et al. (2004), heterogeneous intangible capital generates selection into FDI: only projects associated with sufficiently high levels of intangible capital engage in foreign investment. Second, conditional on investing abroad, projects choose between GFDI and M&A, as in Nocke and Yeaple (2007; 2008). Third, the model incorporates search and matching frictions in the merger market, following the domestic M&A literature (Rhodes-Kropf and Robinson, 2008; David, 2021). I incorporate these mechanisms into a small open economy framework, which allows me to explicitly characterize equilibrium allocations in the host country.

[5] For example, Atalay et al. (2014) demonstrate that firms engage in intangible capital transfer rather than intra-firm trade using data on US multi-plant firms. Ramondo et al. (2016) also show that few foreign affiliates engaged in trade with their parent firms using data on US multinationals.

2.1 Basic Setup

The model is static and characterizes a steady-state equilibrium. FDI is implemented at the project level, with each project corresponding to a potential foreign operation that produces a differentiated variety in the host country. The analysis focuses on the measure of foreign and local projects that are active in equilibrium, rather than on projects' dynamic entry decisions over time.[6]

2.1.1 Production Structure

There are two types of goods in both the source and host countries: differentiated goods and a homogeneous good. Differentiated varieties are produced under monopolistic competition and are non-traded across countries. The homogeneous good is produced under perfect competition using one unit of labor per unit of output and is freely traded between the two countries.

Differentiated varieties correspond to foreign projects in the source country. Because differentiated varieties are non-tradable, a foreign project that serves the host market must be implemented as a production project in the host country.[7] Before foreign entry, differentiated varieties in the host country correspond to local projects. After foreign projects are implemented in the host country, the set of active host-country varieties consists of foreign-owned projects and local projects. Profits from foreign-owned projects are repatriated to the source country through trade in the homogeneous good. Profits from local projects are distributed to host-country households.

Foreign Projects. A foreign project i corresponds to a differentiated variety in the source country. Production follows a Cobb-Douglas technology: $y_i = \tilde{Z} K_i^{\alpha} \ell_i^{\beta}$, where $\tilde{Z}$ is productivity, K_i is the intangible capital of the firm associated with project i at the time the project is undertaken, and ℓ_i is labor input. Intangible capital represents organizational and knowledge-based assets, such as brands, patents, managerial know-how, and supply-chain networks. The level of intangible capital $K_i \in (0, \infty)$ is drawn from a distribution $G(K)$.[8] For simplicity, productivity (e.g., managerial capability) is assumed to be homogeneous at

[6] Firms receive foreign-project opportunities over time, and each opportunity expires with an exogenous probability, so that at any instant a constant mass of foreign-project opportunities faces the entry decision. Because intangible capital is non-rival across projects, a firm deploys the same firm-level intangible stock across whatever opportunities it draws.

[7] This assumption gives foreign entry a market-seeking interpretation. The model abstracts from exports of differentiated final goods and export-platform FDI.

[8] Arrighetti et al. (2014) document substantial firm-level heterogeneity in intangible asset investment and links intangible accumulation to firm-specific factors such as firm size and human capital.

the value $\tilde{Z}$. Each foreign project i can be implemented in the host country, and it incurs costs that depend on its mode of FDI. The mass of available foreign projects (equivalently, source-country differentiated varieties) is constant and equal to M.

Local Projects. A local project j corresponds to a differentiated variety initially available in the host country and follows a Cobb–Douglas production technology: $y_j = \tilde{z}k^{\alpha}\ell_j^{\beta}$, where $\tilde{z}$ is productivity, k is intangible capital, and ℓ_j is labor input. The productivity of local projects is homogeneous and lower than that of foreign projects; specifically, $\tilde{z} \leq \tilde{Z}$. The level of intangible capital is also assumed to be homogeneous across local projects and fixed at k.[9] Market entry of each local variety incurs a fixed entry cost c, and the mass of active local projects N is endogenously determined.

2.1.2 Foreign Direct Investment (FDI)

For each foreign project i, the owner decides whether to undertake production in the host country and, conditional on investing, whether to do so through greenfield investment (GFDI) or cross-border M&A. After FDI decisions are made, three types of projects operate in the host country: GFDI projects g, M&A projects m, and local projects j.[10]

GFDI Projects. For project i, if greenfield entry is chosen, a new project is established in the host country. Through this entry mode, productivity $\tilde{Z}$ and intangible capital K_i are transferred to the host market. A GFDI project operates with the same level of production technology as in the source country.[11] The production function of a GFDI project is

$$y_i^g = \tilde{Z}K_i^{\alpha}\ell_i^{\beta}.$$

Establishing a GFDI project requires paying a fixed cost F, measured in units of the homogeneous good.

M&A Projects. For a given project i, if cross-border M&A is chosen as the entry mode, an existing local project in the host country is acquired. Upon acquisition, the foreign

[9] Target projects' intangible capital is not observable in the available datasets. While this assumption simplifies the analysis, it may raise concerns about whether the lack of heterogeneity affects the model's implications. Appendix A.1 addresses this concern by examining cases where local intangibles vary, and shows that the model's qualitative predictions remain unchanged under the homogeneous assumption.

[10] GFDI adds a new host-country variety. M&A does not add a new variety; instead, it converts an existing local variety into a foreign-owned variety. Thus, the acquired local project disappears from the set of stand-alone local projects but remains in the product space as an M&A project.

[11] For instance, Nocke and Yeaple (2008) and McGrattan and Prescott (2009) suggest that a subsidiary of a multinational operates with the same productivity as its parent firm.

project gains access to the target's intangibles k. The merged project inherits the acquirer's productivity $\tilde{Z}$, as well as both the target's and acquirer's intangible capital, k and K_i. This assumption aligns with the evidence that acquired firms often experience productivity improvements (Schoar, 2002; Li, 2013; Dimopoulos and Sacchetto, 2017), and the post-merger performance depends on the pre-merger attributes of both parties (Guadalupe et al., 2012; David, 2021). The production function of an M&A project is

$$y_i^m = \tilde{Z}(k + \eta_i K_i)^\alpha \ell_i^\beta,$$

where $\eta_i \in (0, 1)$ is an integration-quality draw that governs the effective transfer of foreign intangible capital. The draws η_i are independent and identically distributed (i.i.d.) across matches and independent of the foreign project's intangible capital K_i. Let $H(\eta)$ denote the cumulative distribution function of η_i.

The formulation of the production function reflects a conceptual distinction between technology (Z and z) and intangible capital (K and k). Technology is *non-additive*: when two projects are combined through M&A, the superior managerial practice from one side typically dominates. In contrast, intangible capital is *additive*. Distinct intangible assets—such as patents, trademarks, supply chain relationships, and brand equity—can retain their independent value and jointly contribute to the output of the merged project. For example, a foreign project's proprietary brand can complement a local project's market knowledge or distribution networks, which generates a cumulative gain in production. This distinction highlights that, unlike technology, different components of intangible capital can coexist and contribute jointly, rather than substituting for one another.

Only a fraction η_i of the foreign project's intangible capital remains effective after post-merger integration. Overlapping operations may reduce the effective transfer of the acquirer's resources, while the merged project may continue to rely on the local project's brand, networks, or market-specific capabilities.[12] The realization of η_i occurs after a match is formed. This reflects the fact that the success of post-merger integration is uncertain ex ante and becomes known only after the M&A is initiated.

[12] Appendix A.2 provides anecdotal examples of the use of local intangible assets k and the incomplete integration of acquirer intangible assets ηK after M&A. Unlike in Nocke and Yeaple (2008), the post-merger asset base in my model is specified as the composite intangible input $k + \eta K$. In the Renault–Nissan Alliance, Renault vehicles were sold through Nissan's dealer network in Japan, while some technical assets were not fully integrated (Renault-Nissan Alliance, 2003; Segrestin, 2005). The AB InBev–Grupo Modelo case provides another example: after the acquisition, AB InBev introduced only one additional AB InBev brand into Mexico rather than redeploying its full brand portfolio (Alviarez et al., 2025).

2.1.3 Merger Market

If project i enters the host country via cross-border M&A, it must find a local partner. Conditional on searching, each foreign project attempts to match with a local project in the host country. Searching incurs a fixed cost $\psi > 0$ per project. After a successful match, the foreign project pays an acquisition price V to acquire the local project. Both payments are made in units of the homogeneous good.

The rate at which a searching foreign project matches with a local project is determined by a matching technology. The probability that a searching foreign project finds an M&A partner is given by

$$\mu(n, N) = \left(\frac{1}{1 + (n/N)^{\rho}}\right)^{\frac{1}{\rho}} \in (0, 1), \tag{1}$$

where n denotes the number of searching foreign projects and N the number of active local projects, and $\rho > 0$.[13] When n foreign projects search, μn of them successfully match with local projects, and therefore μn local projects are acquired. Assume that the number of local projects N is sufficiently large so that $N > \mu n$. Because $\partial\mu(n, N)/\partial n < 0$, the matching probability decreases as more foreign projects search (i.e., search congestion occurs).

2.1.4 Households

There is a representative household in the host country endowed with L units of labor. Aggregate income I consists of wage payment wL, profits generated by local projects, and acquisition transfers V. A fraction γ of aggregate income is spent on differentiated varieties y_ω, while the remaining share, $1 - \gamma$, is allocated to a homogeneous good y_0. Each differentiated variety ω is supplied by one of three types of projects operating in the host country: GFDI projects, M&A projects, or local projects. The household's utility function is

$$u = (1 - \gamma) \log y_0 + \gamma \log \left(\int_{\Omega} y_\omega^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \tag{2}$$

where $\sigma > 1$ is the elasticity of substitution across varieties, and Ω denotes the set of all active projects in the host country, $\omega \in \Omega$.

[13] The number of matches is denoted as $v(N, n) = \frac{Nn}{(N^{\rho}+n^{\rho})^{1/\rho}}$, following Den Haan et al. (2000) and Coşar et al. (2016). Unlike a Cobb-Douglas matching function, this functional form ensures that matching probabilities remain between 0 and 1.

2.1.5 Timing

There are four stages in the model:

Stage 1: Local projects enter the host country market.

Stage 2: For each foreign project, a decision is made whether to invest in the host country and, conditional on investing, whether to search for an M&A partner, or invest via GFDI without searching.

Stage 3: Searching foreign projects are matched with local projects. Matched projects decide whether to proceed with M&A, while unmatched projects enter via GFDI if profitable. Projects that choose GFDI without searching are implemented directly. After FDI, three types of projects exist in the host country: GFDI projects, M&A projects, and non-merged local projects.

Stage 4: All active projects in the host country hire workers, produce, and generate profits. Households consume.

2.2 Model Solution

To solve the model, I start with the final stage and work backwards.

2.2.1 Profit Maximization (Stage 4)

In Stage 4, each active project in the host country maximizes profits. The CES demand function is derived from household preferences. For a differentiated variety y_ω,

$$y_\omega = A p_\omega^{-\sigma}, \tag{3}$$

where $A \equiv \gamma I P^{\sigma-1}$ represents the demand level in the host country. The demand level A depends on the aggregate income I and the price index for differentiated varieties $P = \left[\int_\Omega p_\omega^{1-\sigma} d\omega\right]^{1/(1-\sigma)}$.

The profits for each type of project are:

$$\begin{cases} \pi_i^g(K_i, B) \ = BZK_i - F & \text{for GFDI projects,} \\ \pi_i^m(K_i, \eta_i, B) = BZ(k + \eta_i K_i) - V & \text{for M\&A projects,} \\ \pi_j(B) \ = Bzk & \text{for non-acquired local projects,} \end{cases} \tag{4}$$

where $B \equiv \frac{\alpha}{\beta}\left[\frac{\beta(\sigma-1)}{\sigma}A^{1/\sigma}\right]^{\frac{\sigma}{\alpha(\sigma-1)}}$.[14] The host-country wage w is normalized to one. For notational simplicity, let $Z \equiv \tilde{Z}^{1/\alpha}$ and $z \equiv \tilde{z}^{1/\alpha}$. I assume that $\alpha = \sigma/(\sigma-1) - \beta$ with $0 < \beta \leq 1$. This assumption holds without loss of generality because the level of intangibles K is exogenously given. Its unit can always be rescaled through a monotonic transformation to ensure that α satisfies this relationship.[15]

2.2.2 Gain from Mergers (Stage 3)

In Stage 3, for each project that has successfully matched with a local partner, the project is evaluated for implementation as M&A. For projects that either did not search or searched but failed to match, the choice is between GFDI and no investment. When making these decisions, the general equilibrium term B is taken as given. For notational simplicity, I omit the dependence on B in the profit functions (e.g., $\pi_i^g(K_i, B)$ is written as $\pi_i^g(K_i)$).

Profit for a Non-matched Project. Consider a project that searched but failed to match with a local partner. The project is implemented via GFDI only if it yields positive profit.[16] The payoff from GFDI entry is given by

$$\tilde{\pi}_i^g(K_i) = 1\{K_i \geq \tilde{K}\}\pi_i^g(K_i), \tag{5}$$

where $\tilde{K}$ denotes the threshold level of intangible capital such that a GFDI project earns zero profit. Formally, $\tilde{K}$ is defined by $\pi_i^g(\tilde{K}) = 0$; equivalently, $\tilde{K} = \frac{F}{BZ}$. Projects with $K_i \geq \tilde{K}$ undertake GFDI, whereas those with $K_i < \tilde{K}$ do not invest.

Combined Gain from Merger. The total surplus from a merger for project i is given by

$$\begin{aligned}\Sigma(K_i, \eta_i) &= [\pi_i^m(K_i, \eta_i) - \tilde{\pi}_i^g(K_i)] + [V(K_i, \eta_i) - \pi_j], \\ &= BZ(k + \eta_i K_i) - 1\{K_i \geq \tilde{K}\}[BZK_i - F] - Bzk. \end{aligned}\tag{6}$$

The first bracketed term represents the foreign project's net gain from merging: its

[14] Although productivity levels Z and z are assumed to be homogeneous, introducing productivity heterogeneity would generate complementarity between productivity and intangible capital (i.e., $\frac{\partial^2 \pi}{\partial Z \partial K} > 0$), as in Nocke and Yeaple (2008).

[15] The distribution $G(K)$ refers to the post-transformed values of K. This normalization is innocuous only when K is exogenously given. If foreign projects instead choose K optimally through investment, the normalization would no longer be without loss of generality.

[16] Case-study evidence suggests that some firms consider GFDI as an alternative when acquisition opportunities are not suitable. For example, Vita Holdings Limited, a UK-based chemical company, initially sought a local partner in Poland but, after failing to find a suitable target, chose greenfield investment instead (Estrin et al., 1997).

profit as a merged project $\pi_i^m(K_i, \eta_i)$ relative to its outside option $\tilde{\pi}_i^g(K_i)$, which is either GFDI or no investment. The second bracketed term captures the local project's net gain: it receives the acquisition price $V(K_i, \eta_i)$ if acquired, compared to its standalone profit π_j if it remains independent. Since the price $V(K_i, \eta_i)$ is a transfer from the foreign project to the local project, it cancels out in the total surplus. The second line provides the simplified expression after substituting equations (4) and (5).

Acquisition Price. If the merger takes place, the acquisition price, $V(K_i, \eta_i)$, is determined through Nash bargaining over the total merger surplus in equation (6). The price is the sum of the local project's standalone profit, π_j, and its bargaining share χ of the total surplus $\Sigma(K_i, \eta_i)$:

$$V(K_i, \eta_i) = \pi_j + \chi \Sigma(K_i, \eta_i), \tag{7}$$

where $\chi \in (0, 1)$ is the local project's bargaining power.

For a matched project i, M&A is implemented only if the individual gain from merging, $(1-\chi)\Sigma(K_i, \eta_i)$, is positive. This gain accounts for the difference between the merged profit and the foreign project's outside option—either GFDI or no investment. Since $\chi \in (0, 1)$, the merger takes place if $\Sigma(K_i, \eta_i) > 0$.

Trade-off for Investing Projects. The total merger surplus takes two forms because the outside option of a foreign project differs depending on its level of intangible capital:

$$\Sigma(K_i, \eta_i) = \begin{cases} \Sigma_1(K_i, \eta_i) \equiv B[(Z-z)k + Z\eta_i K_i] & \text{for } K_i \in (0, \tilde{K}), \\ \Sigma_2(K_i, \eta_i) \equiv B[(Z-z)k - Z(1-\eta_i)K_i] + F & \text{for } K_i \in [\tilde{K}, \infty), \end{cases} \tag{8}$$

where the surplus functions satisfy $\Sigma_1(\tilde{K}, \eta_i) = \Sigma_2(\tilde{K}, \eta_i)$ for any given η_i.

For projects with $K_i < \tilde{K}$, GFDI yields zero profit and thus the only profitable entry mode is M&A. In this region, the merger surplus $\Sigma_1(K_i, \eta_i)$ is increasing in K_i: a larger intangible stock leads to greater gains from transferring it to the merged project relative to the outside option of not investing.

By contrast, foreign projects with $K_i \geq \tilde{K}$ have GFDI as their outside option. The surplus function $\Sigma_2(K_i, \eta_i)$ reflects a trade-off between M&A and GFDI. On the one hand, M&A enables the project to leverage the productivity gap between foreign and local projects $(Z - z)$ and upgrade local intangibles k. On the other hand, M&A entails losing a fraction $(1 - \eta_i)$ of the project's own intangible capital during the integration process. Because $\eta_i \in (0, 1)$, $\Sigma_2(K_i, \eta_i)$ is decreasing in K_i. This implies that projects undertaken with lower intangible capital have a stronger incentive to acquire local intangible assets through M&A,

whereas projects with higher intangible capital prefer GFDI.

Critical Rate of Integration. Upon matching, a realization of integration quality η_i is drawn for the potential M&A project. Any matched project with $K_i \in (0, \tilde{K})$ proceeds with M&A because $\Sigma_1(K_i, \eta_i) > 0$ for all η_i. For a matched project with $K_i \in [\tilde{K}, \infty)$, GFDI becomes a feasible outside option. In this case, a matched project proceeds with M&A only if the realized integration quality is sufficiently high.

Solving $\Sigma_2(K_i, \eta_i) = 0$ gives the interior cutoff $\eta^\circ(K_i) = 1 - \frac{1}{ZK_i}\left[(Z-z)k + \frac{F}{B}\right]$. Because η_i is supported on $[0,1]$, the relevant cutoff is the truncated value $\bar{\eta}(K_i) = \max\{0, \eta^\circ(K_i)\}$ for $K_i \in [\tilde{K}, \infty)$. Define $K_\eta \equiv \frac{1}{Z}\left[(Z-z)k + \frac{F}{B}\right]$, which solves $\eta^\circ(K_\eta) = 0$; equivalently, $\eta^\circ(K_i) > 0$ if and only if $K_i > K_\eta$. Hence, $\bar{\eta}(K_i) = 0$ for $K_i \in [\tilde{K}, K_\eta)$, so any matched project in this interval proceeds with M&A. For $K_i \in [K_\eta, \infty)$, the realized integration quality affects the post-match choice.

A matched project is implemented as M&A if and only if the realized η_i is greater than or equal to the following effective cutoff:

$$\bar{\eta}(K_i) = \begin{cases} 0 & \text{for } K_i \in (0, K_\eta), \\ 1 - \frac{1}{ZK_i}\left[(Z-z)k + \frac{F}{B}\right] & \text{for } K_i \in [K_\eta, \infty), \end{cases} \tag{9}$$

where $K_\eta = \frac{1}{Z}\left[(Z-z)k + \frac{F}{B}\right] = \tilde{K} + \frac{Z-z}{Z}k$. The effective cutoff $\bar{\eta}(K_i)$ is weakly increasing in K_i: it is flat at zero for $K_i < K_\eta$ and strictly increasing for $K_i \geq K_\eta$. Projects with greater intangible capital obtain smaller marginal gains from acquiring local intangible capital through M&A relative to the GFDI outside option, and therefore require a higher realized level of integration quality η_i for the merger to be optimal.[17]

2.2.3 Search Decision (Stage 2)

In Stage 2, each foreign project decides whether to search for a local partner. Because the realization of η_i is not observed prior to searching, the decision is based on the expected merger surplus with respect to the distribution $H(\eta)$. Each foreign project takes the aggregate demand shifter B, the mass of active local projects N, and the matching probability $\mu(n, N)$ as given. Since each foreign project is infinitesimal, it does not internalize the effect of its own search decision on the aggregate mass of searchers n. I write μ rather than $\mu(n, N)$

[17] One interpretation of the stricter integration requirement is that a project associated with greater intangible capital, especially knowledge capital accumulated through R&D, relies more heavily on firm-specific know-how and production processes, which may be harder to integrate with an existing local operation through M&A.

throughout this subsection when there is no ambiguity.

Search Condition. A foreign project participates in the merger market if the following condition is satisfied:

$$\mu \mathbb{E}_\eta[\max\{\pi_i^m(K_i,\eta), \tilde{\pi}_i^g(K_i)\}] + (1-\mu)\tilde{\pi}_i^g(K_i) - \psi \geq \tilde{\pi}_i^g(K_i), \tag{10}$$

that is, the expected net profit from searching (left-hand side) must be greater than or equal to the outside option of not searching (right-hand side), which is either no investment or GFDI. If a foreign project searches, it matches with a local project with probability μ and remains unmatched with probability $1-\mu$. $\mathbb{E}_\eta[\cdot]$ denotes the expectation over the distribution of η.

Using the expressions above, the search condition (10) can be written as:

$$\mu(1-\chi)\bar{\Sigma}(K_i) \geq \psi, \tag{11}$$

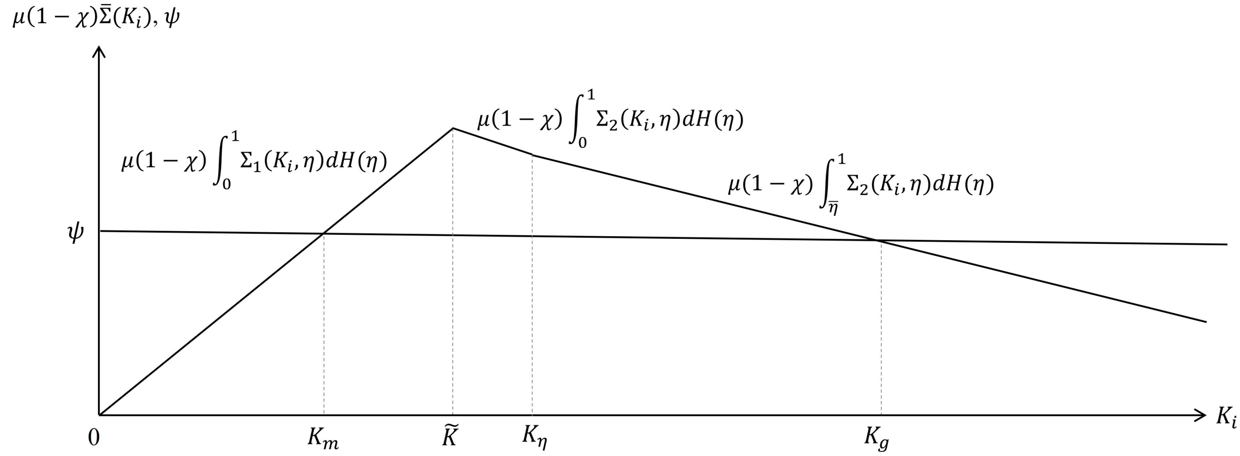
where

$$\bar{\Sigma}(K_i) = \begin{cases} \int_0^1 \Sigma_1(K_i,\eta)dH(\eta) & \text{for } K_i \in (0,\tilde{K}), \\ \int_0^1 \Sigma_2(K_i,\eta)dH(\eta) & \text{for } K_i \in [\tilde{K}, K_\eta), \\ \int_{\bar{\eta}(K_i)}^1 \Sigma_2(K_i,\eta)dH(\eta) & \text{for } K_i \in [K_\eta, \infty). \end{cases}$$

The right-hand side of condition (11) is the fixed cost of search ψ. A foreign project chooses to search if its expected gains from search exceed the cost. The left-hand side represents the expected payoff from entering the merger market: a searching foreign project matches with a local project with probability μ and receives a bargaining share $(1-\chi)$ of expected merger surplus $\bar{\Sigma}(K_i)$. The expected merger surplus $\bar{\Sigma}(K_i)$ takes three forms depending on the project's intangible capital. For projects with $K_i \in (0,\tilde{K})$, any successful match is implemented via M&A, so the relevant surplus is $\int_0^1 \Sigma_1(K_i,\eta)dH(\eta)$. Unmatched projects do not invest. For projects with $K_i \in [\tilde{K}, K_\eta)$, matched projects proceed with M&A for any realization of η, while unmatched projects undertake GFDI. The relevant surplus is therefore $\int_0^1 \Sigma_2(K_i,\eta)\, dH(\eta)$. Finally, for projects with $K_i \in [K_\eta,\infty)$, matched projects proceed with M&A only if $\eta_i \geq \bar{\eta}$; otherwise, they invest via GFDI. Unmatched projects also make GFDI. Thus, the relevant surplus is $\int_{\bar{\eta}(K_i)}^1 \Sigma_2(K_i,\eta)dH(\eta)$.

The surplus functions in (8) imply a non-monotonic expected gain from search. In particular, $\Sigma_1(K_i,\eta)$ is increasing in $K_i \in (0,\tilde{K})$, whereas $\Sigma_2(K_i,\eta)$ is decreasing in $K_i \in [\tilde{K},\infty)$. As Figure 1 illustrates, this non-monotonicity generates two search thresholds, K_m and K_g. Projects with either low or high intangible capital have lower expected surplus and thus

Figure 1: Three Types of Expected Surplus from Search



Note: The figure plots the expected gain from search, $\mu(1-\chi)\bar{\Sigma}(K_i)$, against the level of intangible capital K_i. The search cost ψ is constant. Projects with $K_i \in [K_m, K_g)$ search, while projects outside this interval do not.

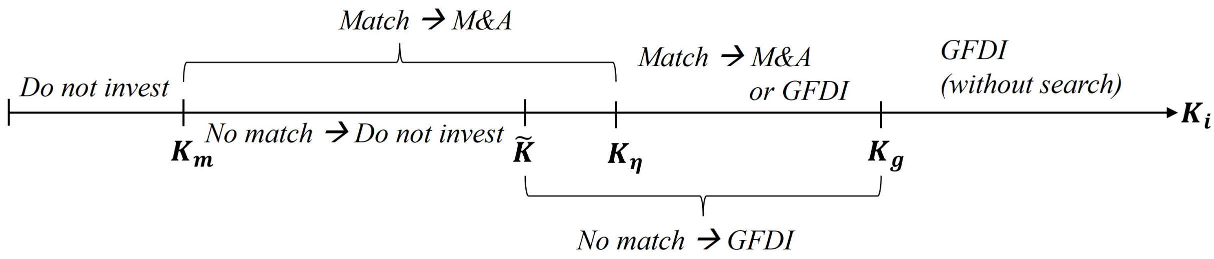
do not find it worthwhile to search (i.e., $\mu(1-\chi)\bar{\Sigma}(K_i) < \psi$). By contrast, projects with intermediate levels of intangible capital find it profitable to search, as their expected surplus exceeds the search cost (i.e., $\mu(1-\chi)\bar{\Sigma}(K_i) > \psi$). This sorting pattern generates two thresholds, K_m and K_g, which define the range of intangible capital over which projects choose to search for M&A partners.

The first threshold K_m is the value of K satisfying $\mu(1-\chi)\int_0^1 \Sigma_1(K_m,\eta)dH(\eta) = \psi$. Projects with $K_i \in (0, K_m)$ do not invest at all, while those with $K_i \in [K_m, \tilde{K})$ choose to search and invest via M&A if matched. In this region, since $K_i < \tilde{K}$, the post-merger draw η_i always satisfies $\eta_i \geq \bar{\eta}(K_i)$. The threshold $\tilde{K}$ denotes the level of intangible capital below which GFDI yields zero profit—that is, unmatched projects with $K_i \in [K_m, \tilde{K})$ do not invest.

The second threshold K_g satisfies $\mu(1-\chi)\int_{\bar{\eta}(K_g)}^1 \Sigma_2(K_g,\eta)dH(\eta) = \psi$. Foreign projects with $K_i \in [\tilde{K}, K_g)$ choose to search, while those with $K_i \in [K_g, \infty)$ invest via GFDI without searching. Among searching projects with $K_i \in [\tilde{K}, K_g)$, those that match and draw a value $\eta_i \geq \bar{\eta}(K_i)$ conduct M&A, while those that either fail to match or draw $\eta_i < \bar{\eta}(K_i)$ undertake GFDI. For expositional clarity, I focus on the case $K_\eta < K_g$, so that there exists a nonempty interval in which the realized integration draw affects the post-match choice. This requires $\mu(1-\chi)\int_0^1 \Sigma_2(K_\eta,\eta)dH(\eta) > \psi$.

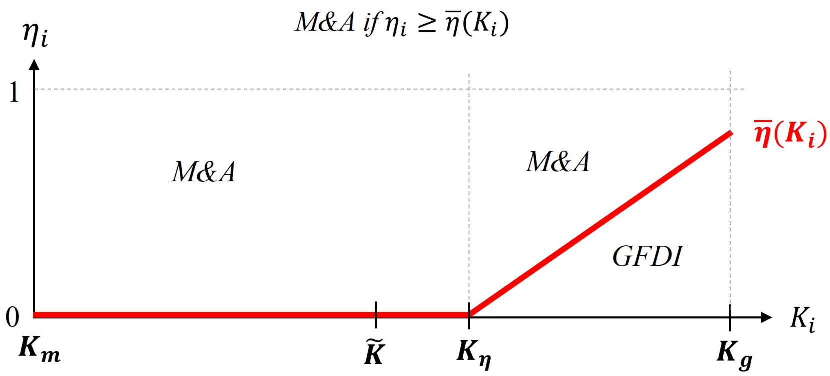
These findings lead to the following proposition:

Figure 2: Mode of Investment by the Level of Intangible Capital



Notes: The figure summarizes a foreign project's FDI decisions as a function of its intangible capital K_i, as described in Proposition 1.

Figure 3: Post-match Decision Based on Realized Integration Quality η_i



Notes: The figure illustrates, for matched projects, the critical integration threshold $\bar{\eta}(K_i)$ that determines whether the matched project is implemented via M&A. Only projects with $K_i \in [K_m, K_g)$ undertake search; among them, M&A occurs if the realized η_i exceeds $\bar{\eta}(K_i)$.

Proposition 1 *A foreign project i chooses among the following investment strategies depending on its level of intangible capital K_i:*

(i) If $K_i \in (0, K_m)$, no FDI occurs.

(ii) If $K_i \in [K_m, \tilde{K})$, the project searches for a local partner. If matched, it is implemented in the host country via M&A for any realization of η_i; otherwise, no investment occurs.

(iii) If $K_i \in [\tilde{K}, K_\eta)$, the project searches for a local partner. If matched, it is implemented in the host country via M&A for any realization of η_i; otherwise, it is implemented via GFDI.

(iv) If $K_i \in [K_\eta, K_g)$, the project searches for a local partner. If matched, it is implemented in the host country via M&A if $\eta_i \geq \bar{\eta}(K_i)$; otherwise, it is implemented via GFDI.

(v) If $K_i \in [K_g, \infty)$, the project is implemented in the host country via GFDI without

searching.

The threshold levels of intangible capital satisfy:

$$\mu(1-\chi)\int_0^1 \Sigma_1(K_m,\eta)dH(\eta) = \psi,$$
$$\tilde{K} = \frac{F}{BZ},$$
$$K_\eta = \tilde{K} + \frac{Z-z}{Z}k, \quad \textit{and}$$
$$\mu(1-\chi)\int_{\bar{\eta}(K_g)}^1 \Sigma_2(K_g,\eta)dH(\eta) = \psi.$$

Let $S_1(K) \equiv \mu(1-\chi)\int_0^1 \Sigma_1(K,\eta)dH(\eta)$ *and* $S_2(K) \equiv \mu(1-\chi)\int_0^1 \Sigma_2(K,\eta)dH(\eta)$*. The above conditions hold under the assumptions that*

$$\lim_{K\to 0} S_1(K) < \psi, \qquad S_1(\tilde{K}) = S_2(\tilde{K}) > \psi, \qquad S_2(K_\eta) > \psi.$$

The model relies on the three assumptions stated above: some projects do not undertake FDI, some projects find it profitable to search, and the realized draw η_i affects the post-match integration decision. Figure 2 illustrates the FDI strategy as a function of intangible capital, highlighting the search decision and corresponding investment outcomes. Once a searching project matches with a local project, it draws η_i and proceeds with M&A only if $\eta_i \geq \bar{\eta}(K_i)$. This post-match decision rule and the critical integration threshold $\bar{\eta}(K_i)$ are illustrated in Figure 3.

Overall, conditional on conducting FDI, projects with larger intangible capital are more likely to choose GFDI, whereas projects with intermediate levels of intangible capital are more likely to choose M&A.

Share of M&A Projects. Intangible capital of foreign projects is distributed according to a cumulative distribution function $G(K)$. In equilibrium, a fraction $G(K_g)-G(K_m)$ of foreign projects search, and a fraction $1-G(K_g)$ are implemented via GFDI without searching. The total mass of searching foreign projects is therefore $n = M\big(G(K_g)-G(K_m)\big)$, which affects the matching probability $\mu(n,N)$.

Each searching project matches with a local project with probability $\mu(n,N)$. Conditional on matching, M&A is implemented only if the realized $\eta_i \geq \bar{\eta}(K_i)$. The share of M&A

projects is then given by:

$$\Phi_m = \mu(n, N) \int_{K_m}^{K_g} [1 - H(\bar{\eta}(K))] \, dG(K). \tag{12}$$

The expression for the share of GFDI projects is provided in Appendix A.3.

2.2.4 Entry of Local Projects (Stage 1)

There are ex-ante identical potential local projects. Each local project must incur a fixed entry cost c. Out of the total number of local projects N, $\Phi_m M$ are acquired through M&A.

In equilibrium, free entry implies that the expected value of entry equals the cost:

$$\phi \, \mathbb{E}[V(K, \eta) \mid \text{acquired}] + (1 - \phi) \, \pi_j = c, \tag{13}$$

where $\phi = (\Phi_m M)/N$ is the probability that a local project is acquired.

The first term represents the expected payment conditional on being acquired: with probability ϕ, a local project is matched with a searching foreign project and receives the expected acquisition price $\mathbb{E}[V \mid \text{acquired}]$. The second term captures the profit from remaining independent: with probability $1 - \phi$, no acquisition occurs and the project earns standalone profit π_j.

Conditional on acquisition, the expected acquisition price is

$$\mathbb{E}[V(K, \eta) \mid \text{acquired}] = \pi_j + \chi \, \frac{\int_{K_m}^{K_g} \int_{\bar{\eta}(K)}^{1} \Sigma(K, \eta) dH(\eta) dG(K)}{\int_{K_m}^{K_g} [1 - H(\bar{\eta}(K))] \, dG(K)}. \tag{14}$$

Each foreign project that potentially matches with a local project has intangible capital K drawn from distribution $G(K)$ and receives an independent draw of match quality η from distribution $H(\eta)$. In the numerator, the double integral averages the surplus $\Sigma(K_i, \eta_i)$ over all matches with $K_i \in [K_m, K_g)$ and over realizations $\eta_i \geq \bar{\eta}(K_i)$ for which M&A is implemented. The denominator corresponds to the probability that a match leads to a merger. The expected price satisfies $\mathbb{E}[V(K, \eta) \mid \text{acquired}] > \pi_j$ because $\Sigma(K_i, \eta_i) \geq 0$ for all merging pairs. Therefore, the possibility of being acquired raises the expected return to local project creation by adding an acquisition option value to the free-entry condition.[18]

[18]Blonigen et al. (2014) study a dynamic mechanism of cross-border M&A in which domestic firms make sunk investments and later become attractive acquisition targets after productivity shocks. The present static model abstracts from post-entry productivity shocks, but it contains a related option-value channel: because the expected acquisition payment is included in the free-entry condition, the possibility of acquisition

2.3 Model Implications

The model delivers the following two testable implications regarding project-level FDI mode choices and the role of the search and matching environment in cross-border acquisitions.

(i) Conditional on investing abroad, projects associated with higher levels of intangible capital are more likely to be undertaken through greenfield investment (GFDI) rather than cross-border M&A.

(ii) Because cross-border M&A requires successful matching with a local partner, the relative attractiveness of M&A depends on the matching environment in the host country. In particular, a higher matching probability μ or lower search cost ψ increases the expected surplus from search and shifts projects toward M&A relative to GFDI.

Section 3 introduces the data, and Section 4 examines these implications empirically. Because the empirical analysis relies only on the model's mode-choice predictions, I defer the full equilibrium system to the quantitative analysis in Section 5.

3 Data

I use two project-level datasets that identify US firms' foreign direct investment entry modes: greenfield foreign direct investment (GFDI) projects from fDi Markets and cross-border M&A deals from SDC Platinum. These datasets allow me to observe, for each foreign-investment episode, whether a US firm enters a foreign market through GFDI or M&A. Because these FDI project datasets do not contain firm-level financial information, I match investing firms to Compustat and use Compustat to measure parent-firm characteristics such as intangible capital, sales, and value added per worker. Thus, the estimation sample is constructed from observed FDI deals, while Compustat is used to append firm characteristics for the subset of investing firms that can be matched to publicly listed US parents.

In the estimation sample, the entry mode is observed at the parent-firm–destination-country–affiliate-industry level. The two-country environment of the model should be interpreted as a building block for a foreign project's entry-mode decision in a generic host-country market. In the empirical analysis, I apply this framework to many host-country and affiliate-industry markets by treating each parent-firm–destination-country–affiliate-industry cell as

raises the ex-ante value of creating a local project. In richer dynamic environments, this channel may be particularly relevant for firm entry in economies with low domestic intangible capital.

one realization of this project-level decision. Appendix B provides additional details on the data sources and the estimation sample.

3.1 FDI Project Data

I source GFDI projects and cross-border M&A deals from the following two databases, covering the period from 2003 to 2018.

GFDI Projects. The greenfield investment data come from the Financial Times' fDi Markets database. This database provides publicly announced cross-border greenfield investment projects and is widely used in recent work on multinational activity.[19] UNCTAD also relies on this source in its annual *World Investment Reports*. Since 2003, the database has recorded cross-border physical investments in new projects, expansions of existing projects, and joint ventures. In this paper, I focus exclusively on new investment projects made by US parent firms, that is, firms headquartered in the US.[20] For each project, the database reports parent-company information and project-level characteristics such as the destination country, project date, and industry sector.

Cross-border M&A Deals. SDC Platinum, produced by Thomson Reuters, records both domestic and cross-border M&A deals worldwide. From this database, I extract completed cross-border deals involving US ultimate acquirers, that is, parent firms headquartered in the United States.[21] I further restrict attention to deals in which the acquirer obtains more than 10% ownership. The 10% threshold is standard in the FDI literature and is commonly used to determine whether the acquiring firm exercises control over the target.[22] SDC reports both announcement and completion dates; to align the timing of M&A deals with the initiation dates of greenfield projects in fDi Markets, I use the announcement date in the main analysis.

3.2 Firm Financial Data

I use Compustat to measure parent-firm characteristics for the subset of investing firms that can be linked to publicly listed US parent firms. The main variable of interest in the empirical

[19] Recent examples using fDi Markets include Davies et al. (2018), Alfaro and Chor (2023), Gopinath et al. (2025), and Kim and Lee (2026).

[20] I do not use the data on expansions of existing projects or joint ventures. These account for around 10% and less than 1% of the investment made by US firms, respectively.

[21] I exclude deals involving investment funds such as hedge funds and sovereign wealth funds, as these acquisitions are more likely to reflect financial or speculative motives rather than entry into a foreign business.

[22] For example, the Bureau of Economic Analysis (BEA) defines foreign affiliates as overseas business entities in which US firms own or control at least 10% of the voting shares. Acquisitions with less than 10% ownership account for only 3% of the completed cross-border deals made by US ultimate acquirers.

analysis is the parent firm's stock of internally generated intangible capital. I first define this measure and then explain how it is constructed. I also briefly describe the other firm financial variables. All financial variables are expressed in real terms using the Consumer Price Index for All Urban Consumers (CPI-U).

3.2.1 Intangible Capital

Intangible capital consists of knowledge- and information-based non-physical assets—such as patents, brands, data, organizational know-how, and firm-specific capabilities—that create future economic benefits. Because such assets are typically developed internally and generally do not appear on the balance sheet, they are difficult to measure directly (Haskel and Westlake, 2017). In this paper, I focus on internally generated intangible capital, which is defined as the sum of *knowledge capital* and *organizational capital*.[23] Knowledge capital captures intangible assets generated through a firm's research and development (R&D) activities, while organizational capital includes human capital, branding, customer relationships, and distribution systems (Peters and Taylor, 2017).

A growing literature measures firm-level intangible capital using Compustat data and treats R&D expenditures and a fraction of selling, general, and administrative (SG&A) expenses as inputs into the accumulation of intangible assets.[24] R&D expenditures are interpreted as investment in knowledge capital, while part of SG&A is interpreted as investment in organizational capital. SG&A includes expenditures such as software, employee training, brand advertising, payments to consultants, and the development of supply and distribution networks. Because firms do not separately report each component of SG&A, the literature typically treats only a fraction of SG&A as investment in organizational capital. While SG&A is not a perfect measure of organizational investment, it remains informative because there is no direct operational measure of organizational capital. Related work also shows that accounting-based measures of internally generated intangible capital are strongly correlated with more direct indicators of intangible assets, such as patent valuations, trademark counts, and measures of brand quality (Ewens et al., 2025). A more detailed discussion of the interpretation and validation of the intangible-capital measure is provided in Appendix

[23] There are two types of intangible capital: internally generated intangible capital and externally generated intangible capital. Compustat reports balance-sheet intangible assets, but these primarily capture externally acquired intangible assets, including goodwill, rather than internally generated ones (Corrado et al., 2022). In this paper, I focus on off-balance sheet internally generated intangible capital, which reflects actual investment in intangible assets without the markups associated with acquisitions, such as goodwill.

[24] In addition to Peters and Taylor (2017) and Ewens et al. (2025), the recent literature includes Eisfeldt and Papanikolaou (2013), Belo et al. (2014), Eisfeldt and Papanikolaou (2014), Gourio and Rudanko (2014), Falato et al. (2022).

Section B.3.

3.2.2 Construction of Intangible Capital Stocks

I construct intangible capital stocks following Ewens et al. (2025), who estimate capitalization parameters for intangible capital using market prices, building on Peters and Taylor (2017). The share of SG&A expense and the depreciation rate of knowledge capital vary across industries. On average, 23% of SG&A spending is treated as investment in organizational capital, and the depreciation rate of knowledge capital is 42%. The depreciation rate of organizational capital is set at 20%. Because there is no clear consensus on the relevant parameter choices, I also report robustness using the methodology of Peters and Taylor (2017) and alternative depreciation rates.

Throughout the analysis, I use parent-firm characteristics lagged by one year relative to the investment decision. Accordingly, the FDI sample spans 2003–2018, while the matched Compustat-based firm characteristics span 2002–2017. I construct intangible capital using firms' R&D and SG&A expenditures beginning in 1980 because intangible capital is a stock measure. To avoid constructing the stock from very short expenditure histories, I impose a six-year minimum-history requirement and retain only matched parent firms that are observed in Compustat for at least six years.[25] The resulting measure is time-varying. Appendix Figure B.3 shows that measured intangible capital rises gradually over firms' observed years in the sample window. Although organizational capital is quantitatively larger on average, knowledge capital is not negligible: among observations for which the share is defined, knowledge capital accounts for 34% of total measured intangible capital on average, with a median share of 33%.

3.2.3 Other Financial Variables

In addition to intangible capital, I obtain sales and construct value added per worker following İmrohoroğlu and Tüzel (2014) to control for firm size and productivity. I also extract physical capital, measured as the book value of property, plant, and equipment, and use it in a placebo specification to assess whether the relationship with FDI mode is specific to intangible capital.

[25] Appendix Section C.4 addresses the concern that stock measures constructed from shorter expenditure histories may be noisier by imposing a stricter firm-year history requirement. Under this restriction, a firm-year enters the regression sample only after six years of expenditure history are available.

3.3 Construction of Estimation Sample

I construct the estimation sample by starting from the observed FDI projects in fDi Markets and SDC Platinum and then appending parent-firm characteristics from Compustat. Compustat is used to supplement the FDI project data with financial information for the subset of investing firms that can be matched to publicly listed US parent firms.[26]

I implement the matching procedure in two steps. First, for SDC Platinum deals involving publicly listed firms, I use CUSIP identifiers when available. Second, for the remaining firms in SDC Platinum and for firms in fDi Markets, I match parent firms to Compustat using company names and headquarters states. I also manually verify firms that changed names over time. The unit of observation is the parent-firm–destination-country–affiliate-industry cell, following Nocke and Yeaple (2008). This level of observation is appropriate because large international firms commonly operate across multiple industries and destinations, and may therefore use different foreign-expansion modes across distinct activities and markets (Bernard et al., 2018; Yeaple, 2023). When a parent firm undertakes multiple investments in the same destination-country–industry cell, I focus on the first observed investment in each market because the empirical question concerns the firm's initial observed mode of entry, rather than subsequent expansions of existing affiliates. This choice is unlikely to discard substantial information about mode choice because mode switching within a market is limited: among observed parent-firm–destination-country–affiliate-industry cells in the estimation sample, only 4.3% exhibit both GFDI and M&A over the sample period, and the first observed mode is highly persistent within a market.[27]

Because Compustat covers only publicly listed firms, some firms may have recorded foreign investments before they first become eligible to enter the regression sample. I therefore exclude firms for which the foreign-investment data record any investment before the first year in which lagged Compustat controls are available. This prevents a later investment from being misclassified as the firm's first observed investment in the estimation sample solely because earlier investments occur before Compustat coverage. The final matched estimation sample contains 2,471 publicly listed US parent firms linked to the FDI project data.

Finally, FDI project data begin in 2003, and so I cannot directly observe all foreign activity undertaken before that date. To address this concern, I use the WRDS company

[26] Thus, the estimation sample is defined by observed foreign investment episodes rather than by the universe of Compustat firms. Appendix Section B.4.2 provides additional details on the matched sample.

[27] For example, 96% of cells whose first observed entry is GFDI are never followed by M&A, and 95% of cells whose first observed entry is M&A are never followed by GFDI, in subsequent foreign investment episodes in the same destination-country–industry cell (Table B.2).

subsidiary data to identify whether a firm had foreign affiliates in a destination country before 2003 or before the firm first appears in Compustat.[28] Since the industry classification of these pre-existing foreign affiliates is not available, I cannot determine whether they operated in the same affiliate industry as the focal investment. I therefore use this information conservatively in robustness checks by excluding destination countries in which the parent firm had prior foreign affiliates.

3.4 Host-Country Characteristics

I augment the matched FDI sample with variables describing destination-country characteristics. I measure the level of development using GDP per capita and market size using population, both from the Penn World Table. GDP per capita is constructed as output-side real GDP at chained PPPs divided by population. I measure openness to trade as the ratio of exports plus imports to GDP, using data from the World Bank database. From the CEPII database, I obtain bilateral distance between the US and each host country, as well as an indicator for whether the host country shares a common language with the US.

To capture target-market conditions in host countries, I also use measures of the number of local firms and the stock of local intangible capital. I obtain the number of local firms with more than 50 employees from the OECD Structural Business Statistics database and local intangible-capital stocks from the EUKLEMS & INTANProd database.[29] These variables are available only for subsets of host countries, industries, and years, but they are useful for examining how the local target environment affects multinationals' FDI mode choices.[30] In the empirical analysis, I normalize both the number of local firms and local intangible capital by population to capture local target-market conditions relative to host-economy size.[31]

[28] The WRDS company subsidiary data report the names and locations of subsidiaries of Compustat firms, based on SEC filings such as Form 10-K.

[29] This measure is conceptually analogous to the Compustat-based intangible-capital measure for US parent firms in that both are stock measures constructed from accumulated intangible investment flows. Details are provided in the discussion of national-account measures of intangible capital in Data Appendix B.3. I use the chain-linked volume measure of total intangible capital stock. The stock values are reported in millions of 2015 national currency, so I convert them into 2015 PPP-adjusted US dollars by dividing by the destination country's 2015 GDP purchasing power parity from the OECD Annual Purchasing Power Parities and Exchange Rates table.

[30] The OECD database reports the number of firms with more than 50 employees for 44 countries by ISIC Rev. 4 industry classification. The EUKLEMS & INTANProd database reports local intangible-capital stocks for 20 countries by NACE Rev. 2 industry classification.

[31] Population provides a neutral scale adjustment and makes these measures comparable to other per-capita host-country characteristics, such as GDP per capita.

Table 1: Summary Statistics

	My data						Nocke & Yeaple		
	All industries			Manufacturing only					
Variable	Mean	S.D.	Obs.	Mean	S.D.	Obs.	Mean	S.D.	Obs.
Panel A: Baseline cell-level sample									
M&A	0.416	0.493	15,544	0.418	0.493	8,787	0.435	0.496	856
Intangible capital	20.295	2.120	15,544	20.540	2.044	8,787	-	-	-
Organizational capital	19.996	2.090	15,543	20.169	2.038	8,786	-	-	-
Knowledge capital	19.477	2.327	9,434	19.499	2.222	6,845	-	-	-
Physical capital	20.585	2.502	15,326	20.922	2.415	8,778	-	-	-
Sales	21.634	2.246	15,544	21.830	2.188	8,787	15.34	1.61	856
Value-added per worker	4.379	0.673	14,900	4.383	0.582	8,489	4.49	0.523	849
Panel B: Destination-country characteristics									
Distance	8.769	0.815	15,402	8.809	0.776	8,775	8.72	0.69	856
Common language	0.376	0.484	15,402	0.344	0.475	8,775	-	-	-
GDP per capita	10.052	0.838	14,881	10.017	0.849	8,490	9.81	0.723	856
Population	17.608	1.646	14,881	17.698	1.680	8,490	16.7	1.38	856
Openness	4.264	0.558	15,213	4.261	0.556	8,646	3.94	0.648	856
Number of local firms/pop	2.625	1.113	5,180	2.498	1.062	2,876	-	-	-
Local intangibles/pop	6.321	1.038	4,259	6.072	0.994	2,434	-	-	-

Note: The unit of observation is at the parent-firm–destination-country–affiliate-industry level. Continuous variables are in logs. Distance is measured in kilometers, and monetary variables are in US dollars. The number of local firms and local intangibles are divided by population. Observation counts vary across variables because some firm characteristics require additional accounting information, and some country variables are unavailable for all destinations. For financial variables, the reported values come from the one-year-lagged Compustat match to each FDI observation. For the benchmark comparison with Nocke and Yeaple (2008), the reported means of monetary variables are adjusted using the Consumer Price Index for All Urban Consumers (CPI-U).

3.5 Descriptive Patterns in the Matched FDI Sample

I next summarize the main features of the matched FDI sample. Table 1 reports summary statistics for the baseline estimation sample and, as a benchmark, compares them with the BEA-based sample used by Nocke and Yeaple (2008).[32] Although my sample is limited to publicly listed US parent firms and covers 2003–2018 rather than 1994–1998, the share of M&A and the key destination-country characteristics are similar across the two datasets. Unlike Nocke and Yeaple (2008), my data also cover service-sector FDI.[33]

[32] I aggregate the data in a slightly different way from Nocke and Yeaple (2008). For firms with more than one investment in a given country–industry cell, Nocke and Yeaple (2008) classify a firm as using M&A only if all investments made during the sample period are through M&A.

[33] The project databases also report transaction-size measures. I use these measures only descriptively, rather than as a main empirical outcome, because they are difficult to compare directly across entry modes: fDi Markets reports estimated capital-expenditure values for GFDI projects, whereas SDC Platinum re-

In the matched FDI data, I observe both the affiliate-side industry reported in the FDI project data (from either fDi Markets or SDC Platinum) and the parent firm's main industry from Compustat. This allows me to classify each FDI project as either intra- or inter-industry. In 56% of observed parent-firm–destination-country–affiliate-industry cells, the affiliate industry differs from the parent firm's main industry.[34] This pattern highlights the value of organizing the data at the firm-market level, rather than treating foreign expansion as a single firm-wide choice. In the matched sample, the same parent firm often undertakes foreign expansion in different affiliate industries, and its choice between GFDI and M&A can also vary across those industries.

Prior work emphasizes different sources of variation in FDI mode choice. For example, Nocke and Yeaple (2008) emphasize firm-level efficiency, whereas Davies et al. (2018) and World Bank (2018) focus on host-country frictions and industry characteristics, respectively. Consistent with this broader discussion, Appendix Table B.4 reports partial-determination measures showing that variation in entry mode is more strongly associated with parent-firm heterogeneity than with destination-country or affiliate-industry characteristics. This descriptive pattern motivates the paper's focus on parent-firm characteristics in the empirical analysis, while maintaining the parent-firm–destination-country–affiliate-industry cell as the relevant unit of observation. A separate data limitation concerns reported transaction-size measures. These measures are difficult to compare directly across entry modes because fDi Markets reports capital-expenditure values for GFDI projects, whereas SDC Platinum reports acquisition values only for the subset of M&A deals with disclosed deal values.[35]

4 Empirical Evidence of FDI Entry Modes

This section evaluates the model implications that I introduced in Section 2.3 using my FDI data.

4.1 Parent-Firm Intangible Capital and Entry Mode

The first model implication is that, conditional on investing abroad, projects associated with parent firms with greater internally accumulated intangible assets are more likely to

ports acquisition values only for the subset of M&A deals with available deal values. Appendix Figure B.4 summarizes reported transaction size by entry mode and parent-firm intangible intensity.

[34]The inter-industry share is higher for M&A than for GFDI (63% versus 52%).

[35]For this reason, I do not use transaction size as a main empirical outcome. Appendix Figure B.4 provides a descriptive comparison of reported transaction size by entry mode and parent-firm intangible intensity.

choose greenfield foreign direct investment (GFDI) rather than cross-border M&A. I treat each parent-firm–destination-country–affiliate-industry cell as the empirical counterpart of a project-level FDI mode choice in the model. To examine this prediction, I estimate linear probability models at the parent-firm–destination-country–affiliate-industry level. For parent firm i, destination country h, affiliate industry a, and year t, the estimating equation is:

$$\begin{aligned} MA_{i,h,a,t} = \beta_1 \times Intangibles_{i,t-1} + \beta_2 \times Sales_{i,t-1} + \beta_3 \times ValueAdded_{i,t-1} \\ + \lambda_{h,t} + \lambda_{\iota_i,a} + \varepsilon_{i,h,a,t}, \end{aligned} \tag{15}$$

where $MA_{i,h,a,t}$ is an indicator equal to one if the first observed entry by parent firm i in destination country h and affiliate industry a, occurring in year t, is through M&A and zero if it is through GFDI. Here, ι_i denotes the main industry of parent firm i, while a denotes the industry of the foreign affiliate. All explanatory variables are measured at the parent-firm level, and are in logs. The variable $Intangibles_{i,t-1}$ denotes the parent firm's stock of internally generated intangible capital in year $t-1$. Using lagged explanatory variables prevents a potential simultaneity issue between firms' investment decisions and their financial status in the same period.[36] I also control for firm size and productivity using sales $Sales_{i,t-1}$ and value added per worker $ValueAdded_{i,t-1}$.[37] The country-year fixed effects $\lambda_{h,t}$ control for destination-year shocks that affect the relative attractiveness of M&A and GFDI, such as policy changes related to M&A, while the industry-pair fixed effects $\lambda_{\iota_i,a}$ absorb persistent differences across parent-industry–affiliate-industry pairs, including whether the parent and affiliate industries are horizontally or vertically related.

Table 2 presents the results. Column 1 reports the baseline relationship between total intangible capital and the probability of choosing M&A without additional parent-firm controls. The coefficient on intangible capital is negative and statistically significant. Column 2 adds parent-firm sales and value added per worker; the coefficient on total intangible capital remains negative and statistically significant, and its magnitude increases. These results support the model's prediction that projects associated with parent firms with greater internally accumulated intangible assets are more likely to enter through GFDI rather than through cross-border M&A. Column 3 replaces industry-pair fixed effects with industry-pair–year fixed effects to control for shocks specific to an industry pair between an investing firm and

[36] This timing convention follows Spearot (2012), who studies firms' choice between new capital investment and acquisition in the US using Compustat data.

[37] I include intangible capital and sales separately rather than using the ratio $(intangibles/sales)_{i,t-1}$. Using the ratio is equivalent to imposing the restriction that the coefficients on $\log intangibles_{i,t-1}$ and $\log sales_{i,t-1}$ are equal in magnitude and opposite in sign. I do not impose this restriction in the baseline specification.

Table 2: Intangible Capital and FDI Entry Mode

	Total intangibles			Organizational	Knowledge	Physical
Dep. var: M&A indicator	(1)	(2)	(3)	(4)	(5)	(6)
Intangible capital	-0.022***	-0.043***	-0.051***			
	(0.003)	(0.008)	(0.010)			
Organizational capital				-0.032***		
				(0.008)		
Knowledge capital					-0.024***	
					(0.009)	
Physical capital						-0.009
						(0.009)
Sales		0.013	0.022**	0.003	-0.004	-0.017*
		(0.009)	(0.010)	(0.009)	(0.010)	(0.010)
Value-added per worker		0.017	0.015	0.016	0.022*	0.016
		(0.011)	(0.012)	(0.011)	(0.013)	(0.011)
Country × Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry-pair FEs	Yes	Yes	No	Yes	Yes	Yes
Industry-pair × Year FEs	No	No	Yes	No	No	No
Observations	14,653	14,024	11,914	14,024	8,434	13,847
R-squared	0.469	0.477	0.538	0.476	0.472	0.476
RMSE	0.382	0.380	0.380	0.381	0.384	0.381

Notes: The unit of observation is a parent-firm–destination-country–affiliate-industry cell. The dependent variable equals one if the first observed FDI in the cell is an M&A. All parent-firm financial variables are lagged by one year relative to the entry-mode decision. Standard errors, reported in parentheses, are clustered by parent firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

an affiliate firm in year t (such as demand or supply shocks affecting an industry pair, j_i and j). Although this specification drops singleton parent-industry–affiliate-industry–year observations, the coefficient remains negative and statistically significant, with a magnitude close to that in column 2.

Columns 4–6 keep the same baseline controls and fixed effects as column 2 while replacing the main regressor. Columns 4 and 5 replace total intangible capital with organizational capital and knowledge capital, respectively. Both components enter negatively and significantly.[38] These results suggest that the negative relationship between total intangible capital and M&A is not attributable solely to one component of the stock measure. Finally, column 6 reports a placebo specification using physical capital. In contrast to intangible capital, physical capital is not significantly associated with the probability of choosing M&A. This is

[38]The smaller number of observations in column 5 reflects the fact that knowledge capital can be constructed only for firms with sufficient R&D histories and non-missing R&D expenditures.

consistent with the idea that the relevant mechanism in FDI mode choice operates through internally accumulated intangible assets rather than physical capital, which firms can obtain under either GFDI or M&A.

The appendix reports several additional robustness checks. First, Appendix Table C.1 uses Compustat Supplemental information on foreign affiliates to exclude destination countries in which the parent firm had prior foreign presence before 2003 or before first appearing in Compustat. The estimated relationship between intangible capital and entry mode remains negative and statistically significant. Second, Appendix Tables C.2 and C.3 show that the coefficient on parent-firm intangible capital remains negative and statistically significant when intangible capital is constructed using the methodology of Peters and Taylor (2017) and when alternative depreciation rates are used. Third, Appendix Table C.4 considers alternative sample restrictions related to repeated use of entry modes within a parent-firm–destination-country–affiliate-industry cell. The coefficient on parent-firm intangible capital remains negative and statistically significant when I exclude cells that ever exhibit both entry modes and when I restrict the sample to firms that use only one FDI mode over the sample period. Finally, Appendix Table C.5 reports a more conservative robustness check using a stricter firm-year history requirement. Under this restriction, a firm-year enters the regression sample only after six years of expenditure history are available for stock construction. The coefficient on parent-firm intangible capital remains negative and statistically significant across the main specifications. These results indicate that the baseline relationship is not driven by firms with shorter observed expenditure histories.

Overall, the regression evidence is consistent with the first model implication. Among observed foreign-investment projects, those associated with parent firms with greater internally accumulated intangible assets are more likely to enter through GFDI rather than through cross-border M&A. These results also add to the literature on the determinants of firms' FDI mode choices. For example, Nocke and Yeaple (2008) emphasize firm-level productivity in shaping the choice between greenfield investment and M&A. My results suggest that measured intangible capital is an additional determinant of firms' FDI mode decisions, beyond parent-firm size and productivity controls such as sales and value added per worker.

4.2 Host-Country Conditions and FDI Entry Mode

The second model implication is that cross-border M&A becomes relatively more attractive when the expected surplus from search is higher. In the model, this occurs when the matching probability μ is higher or the search cost ψ is lower. In the empirical analysis, I proxy a higher matching probability by a thicker destination-country–affiliate-industry market, measured by the number of local firms, and a lower search cost by variables such as geographic distance and common language. In addition, Appendix A.1 shows that higher local intangible capital raises merger surplus and expands the region of projects for which M&A is attractive. I therefore also examine destination-country–affiliate-industry local intangible assets as an additional target-side determinant of FDI mode choice. Because the empirical question concerns the ex ante choice between GFDI and M&A, the relevant target-side variables are market-level measures of potential targets rather than the characteristics of a realized target firm, which are observed only for M&A transactions and are not defined for GFDI projects.

Because the purpose of this subsection is to examine how host-country conditions shape FDI mode choice, I do not include country-year fixed effects. Instead, I control for observable host-country characteristics. These include bilateral distance, a common-language indicator, GDP per capita, population, and openness to trade. To capture host-market conditions at the destination-country–affiliate-industry–year level, I also include the number of local firms per population and the stock of local intangible capital per population. These host-market variables are available only for subsets of host countries and years, so the specifications using them are estimated on smaller samples.[39] To distinguish the host-country intangible measure from the Compustat-based parent-firm measure used in the previous subsection, I refer to the latter as parent intangibles. All continuous variables are expressed in logs.

Table 3 reports the results.[40] Column 1 begins with broad host-country controls. Consistent with the idea that cross-border acquisitions are more difficult in remote markets where investing firms face higher search and information frictions, distance enters negatively and significantly, while a common language is associated with a higher probability of choosing M&A. This finding is consistent with Davies et al. (2018), who show that geographical and cultural barriers hinder cross-border M&A more than GFDI. Parent-firm intangible capital continues to enter negatively, which indicates that, all else equal, projects associated with more intangible-intensive parents are more likely to choose GFDI rather than M&A.

Column 2 focuses on target-market thickness. Using the number of local firms per popu-

[39]Appendix Table C.6 re-estimates the broad-controls specification on the reduced samples used in the regressions with the number of local firms and local intangibles.

[40]The regression specification is closest to Nocke and Yeaple (2008). Relative to that paper, I use a similar firm-level entry-mode framework but study a different set of covariates. In particular, I introduce parent-firm intangible capital as the central parent-side variable and augment the specification with additional bilateral and destination-country–affiliate-industry variables, including common language, the number of local firms, and local intangible capital. The regressions also differ from Head and Ries (2008) and Davies et al. (2018), which analyze bilateral or origin–destination–sector outcomes rather than the conditional entry-mode choice of observed parent firms.

Table 3: Host-Country Target Availability and FDI Entry Mode

	Broad controls (142 countries)	Local firm counts (44 countries)	Local intangibles (20 countries)		
Dep. var: M&A indicator	(1)	(2)	(3)	(4)	(5)
Parent intangibles	-0.037***	-0.029**	-0.052***	-0.052***	-0.129***
	(0.009)	(0.014)	(0.012)	(0.012)	(0.025)
Distance	-0.072***	-0.078**	-0.584***	-0.540***	-0.537***
	(0.019)	(0.035)	(0.071)	(0.082)	(0.083)
Common language	0.083**	0.040**			
	(0.032)	(0.018)			
Num local firms per capita		0.034***			
		(0.011)			
Local intangibles per capita			0.060***	0.025*	-0.218***
			(0.014)	(0.014)	(0.071)
Parent intang. $\times$ Local intang.					0.012***
					(0.004)
GDP per capita	0.139***	0.266***		0.310***	0.302***
	(0.025)	(0.022)		(0.047)	(0.048)
Industry-pair FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes
Observations	13,772	5,180	4,259	4,259	4,259
R-squared	0.355	0.326	0.279	0.286	0.288
RMSE	0.404	0.424	0.437	0.435	0.434

Notes: The unit of observation is a parent-firm–destination-country–affiliate-industry cell. The dependent variable equals one if the first observed FDI in the cell is an M&A. All columns include openness, parent-firm sales, and value added per worker. Column 1 controls for population, while columns 2–5 do not include population separately because they use per-population destination-industry measures. Common language is omitted from columns 3–5 because the indicator equals one only for the United Kingdom in the reduced-country sample. Standard errors are two-way clustered by parent firm and by the level at which the destination-side regressor varies: destination country in column 1 and destination-country–industry market in columns 2–5. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

lation as a measure of the pool of potential targets, I find that M&A is relatively more likely in destination-country–industry markets with more local firms. This pattern is consistent with the second model implication: a thicker target market raises the matching probability μ, increases the expected surplus from search, and therefore makes M&A relatively more attractive than GFDI. In the reduced 44-country sample, the coefficients on common language and distance remain statistically significant.

Columns 3–5 examine whether the local intangible-capital environment in the destination market affects the choice of FDI entry mode. Column 3 first includes local intangible capital per population without controlling for destination-country GDP per capita. The coefficient is positive and statistically significant, suggesting that M&A is more likely in destination-country–industry markets with richer local intangible assets. Additionally, common language is omitted from the local-intangible specifications because, in the reduced-country sample, the indicator equals one only for the United Kingdom and therefore provides little variation.[41] Column 4 adds destination-country GDP per capita. The coefficient on local intangible capital per population becomes smaller but remains positive, indicating that part of the association reflects broader host-country development, while the local-intangibles measure is not simply a proxy for destination-country income.

Column 5 allows the effect of local intangible assets to vary with parent-firm intangible capital. The interaction term is positive and statistically significant, indicating complementarity: the association between local intangible assets and the probability of choosing M&A becomes more positive for parent firms with higher intangible capital. Appendix Figure C.1 illustrates this interaction by plotting the marginal effect of local intangible assets on the probability of choosing M&A at the 25th, 50th, and 75th percentiles of parent-firm intangible capital. The point estimates are positive and increasing across the parent-firm intangible-capital distribution. The effect is imprecisely estimated at lower percentiles but becomes statistically distinguishable from zero for high-intangible parent firms. This pattern is consistent with Appendix A.1, which shows that higher local intangible capital raises merger surplus and expands the region of projects for which M&A is attractive, especially for parent firms with greater intangible capital.

Taken together, these results support the second model implication. The bilateral distance and common language coefficients are gravity-type correlates consistent with higher search and information frictions. In line with this, cross-border M&A is relatively more likely in destination-country–affiliate-industry markets with a thicker pool of potential targets and lower search frictions. In addition, the model extension (Appendix A.1) suggests that local intangible assets can further raise the attractiveness of M&A, and the interaction results indicate that this role is stronger for parent firms with greater intangible capital.

5 Quantitative Analysis

The empirical analysis documents that parent-firm intangible capital is systematically related to the choice between GFDI and cross-border M&A. I now use the model to quantify the

[41] Including the common-language indicator leaves the coefficients on parent-firm intangible capital and distance negative and statistically significant, and leaves the coefficient on local intangible capital positive and statistically significant.

aggregate implications of this selection mechanism.

5.1 Equilibrium Conditions

This subsection closes the model by defining the domestic equilibrium in the host country. The preceding analysis characterizes foreign firms' entry-mode choices conditional on the aggregate demand shifter B, the mass of local projects N, and the matching probability $\mu(n, N)$. In equilibrium, these objects are jointly determined with the entry-mode cutoffs.

The independent equilibrium objects are $\{K_m, K_g, n, N, B\}$. I first summarize the conditions that characterize these objects and then define the domestic equilibrium.

5.1.1 Foreign Project Cutoff Conditions

Foreign projects choose their entry modes optimally given the four cutoffs defined by Proposition 1: K_m, $\tilde{K}$, K_η, and K_g. First, I focus on the search cutoffs K_m and K_g. Foreign projects with $K_i < K_m$ do not invest in the host country. Foreign projects with $K_i \in [K_m, K_g)$ search for local projects, and foreign projects with $K_i \geq K_g$ enter through GFDI without searching.

Given B and N, the search cutoffs K_m and K_g satisfy:

$$\mu(n, N)(1-\chi)\int_0^1 \Sigma_1(K_m, \eta, B)dH(\eta) = \psi \tag{16}$$

$$\mu(n, N)(1-\chi)\int_{\bar{\eta}(K_g,B)}^1 \Sigma_2(K_g, \eta, B)dH(\eta) = \psi. \tag{17}$$

The two search cutoffs K_m and K_g determine the mass of searching foreign projects n:

$$n = M\left[G(K_g) - G(K_m)\right]. \tag{18}$$

Given a candidate pair $\{B, N\}$, the three equilibrium conditions (16), (17), and (18) determine the conditional search block $\{K_m, K_g, n\}$.

Other cutoffs, $\tilde{K}(B)$ and $K_\eta(B)$, are determined given B. The GFDI zero-profit cutoff is

$$\tilde{K}(B) = \frac{F}{BZ}, \tag{19}$$

and the integration cutoff is

$$K_\eta(B) = \tilde{K}(B) + \frac{Z-z}{Z}k. \tag{20}$$

5.1.2 Local Free Entry Condition

Local projects satisfy the free entry condition defined in Section 2.2.4. Using the M&A share in equation (12) and the expected acquisition price in equation (14), the free entry condition (13) can be written as

$$\pi_j(B) + \chi \frac{\mu(n, N)M}{N} \int_{K_m}^{K_g} \int_{\bar{\eta}(K,B)}^{1} \Sigma(K, \eta, B) dH(\eta) dG(K) = c. \tag{21}$$

This condition equates the expected gross payoff from local entry—the stand-alone profit plus the expected acquisition premium—to the entry cost. Together with the search block and the expenditure condition below, this local free entry condition pins down the mass of local projects N.

5.1.3 Expenditure Condition

Consider expenditure clearing in the market for differentiated varieties. Aggregate income of host-country households I is given by

$$I = L + (N - \Phi^m M)\pi_j(B) + \Phi^m M E[V(K, \eta) \mid \text{acquired}] - Nc.$$

Since the wage is normalized to one, the first term, L, is wage income. The second term is the gross profit generated by stand-alone local projects. Since Φ^m is the share of foreign projects implemented through M&A, defined by equation (12), $\Phi^m M$ is the mass of acquired local projects and $N - \Phi^m M$ is the mass of local projects that remain independent. A local project that is not acquired operates independently and generates the stand-alone profit $\pi_j(B)$. The third term $\Phi^m M E[V(K, \eta) \mid \text{acquired}]$ is the aggregate acquisition payments received by owners of acquired local projects. The final term subtracts the entry costs paid by local project owners. Using the fact that the expected net return from local project ownership is zero, host-country household income reduces to wage income[42]

$$I = L. \tag{22}$$

Because households spend a fraction γ of income I on differentiated varieties, aggregate

[42] Multiplying the free-entry condition (13) by N yields $(N - \Phi^m M)\pi_j(B) + \Phi^m M E[V(K, \eta) \mid \text{acquired}] - Nc = 0$.

expenditure satisfies

$$
\begin{aligned}
\gamma L = \frac{\sigma}{\alpha(\sigma-1)} B \Bigg[& M \int_{K_g}^{\infty} ZK\, dG(K) + [1-\mu(n,N)]\, M \int_{\widetilde{K}(B)}^{K_g} ZK\, dG(K) \\
& + \mu(n,N) M \int_{K_\eta(B)}^{K_g} H(\bar{\eta}(K,B)) ZK\, dG(K) \\
& + \mu(n,N) M \int_{K_m}^{K_g} \int_{\bar{\eta}(K,B)}^{1} Z(k+\eta K)\, dH(\eta) dG(K) \\
& + \left[N - \mu(n,N) M \int_{K_m}^{K_g} [1 - H(\bar{\eta}(K,B))]\, dG(K) \right] zk \Bigg].
\end{aligned} \tag{23}
$$

The left-hand side is aggregate household expenditure on differentiated varieties. The right-hand side is the total revenue of all active differentiated varieties in the host country. The bracketed term sums the effective technology-intangible components of all active varieties: GFDI projects that enter without search, unmatched searching projects that enter through GFDI, matched projects that choose GFDI after a low integration draw, M&A projects, and stand-alone local projects. For example, the revenue of non-searching GFDI projects is $\sigma/[\alpha(\sigma-1)]BM \int_{K_g}^{\infty} ZK\, dG(K)$.

Equation (23) pins down the demand shifter B jointly with other equilibrium objects $\{K_m, K_g, n, N\}$. The detailed derivation is provided in Appendix Section A.4.

5.1.4 Equilibrium

I now define the domestic equilibrium using the conditions stated above.

Definition 1 *Given the scalar parameters $\{Z, z, k, \chi, \sigma, \beta, M, L, \psi, c, \rho, \gamma, F\}$, the distribution G of foreign intangible capital K, and the distribution H of integration quality η, a domestic equilibrium is characterized by the search cutoffs K_m and K_g, the mass of searching foreign projects n, the mass of local projects N, and the aggregate demand shifter B, which satisfy: (i) Lower search cutoff condition* (16)*, (ii) Upper search cutoff condition* (17)*, (iii) Search-market consistency condition* (18)*, (iv) Free entry condition* (21)*, and (v) Expenditure condition* (23)*.*

Given these objects, the cutoffs $\widetilde{K}(B)$ and $K_\eta(B)$, the threshold function $\bar{\eta}(K,B)$, and the entry-mode shares Φ^m and Φ^g are implied.

The expenditure-clearing condition (23) clears the market for differentiated varieties. The implied allocation must also satisfy labor-market feasibility. Labor-market feasibility requires

$$\frac{\beta(\sigma-1)}{\sigma}\gamma L + Nc + \Phi^g MF + L^0 = L.$$

The right-hand side is the fixed labor supply L, while the left-hand side is total labor use in the host country. The first term is labor used in the production of all active differentiated varieties.[43] The second term Nc is the host-country resource cost of local project entry, and the third term $\Phi^g MF$ is the host-country resource cost of GFDI setup.[44] Because Φ^g is the share of foreign projects implemented through GFDI, defined in equation (A.1), $\Phi^g M$ is the mass of GFDI projects. The term L^0 is residual employment in the homogeneous-good sector. I focus on feasible equilibria with $L^0 \geq 0$. I do not impose a separate clearing condition for the homogeneous-good market; given expenditure clearing for differentiated varieties and labor-market feasibility, the homogeneous-good market clears by Walras' law.

5.2 Calibration

I calibrate the model to discipline the margins that determine the allocation of foreign projects across no FDI, GFDI, and cross-border M&A. I use the data described in Section 3. I set the source country as the US and conduct the calibration for three destination samples: all host countries, advanced economies, and non-advanced economies. The classification of destination countries follows the IMF's economy groupings. The separate advanced and non-advanced calibrations allow the model to capture differences in host-country conditions and entry-mode patterns, including the relatively high M&A share in advanced economies and the greater reliance on GFDI in non-advanced economies. For moments based on observed FDI projects, I pool first observed parent-firm–destination-country–affiliate-industry cells over 2003–2018 within each destination sample. I therefore interpret each calibrated equilibrium as describing a destination market over the sample period.

5.2.1 Externally Assigned Parameters

I first assign a set of parameters using the data and values from the literature. Table 4 reports externally assigned and internally calibrated parameters.

Distribution of Foreign Intangible Capital. I assume that the distribution of foreign in-

[43] The derivation is provided in Appendix A.5.

[44] The search cost ψ captures the cost that foreign owners incur when searching for local targets and gathering information for due diligence. I interpret this cost as using source-country resources, so it does not enter the host-country resource constraint.

tangible capital $G(K)$ is log-normal.[45] Specifically, $\log K_i$ is normally distributed with mean $-\theta^2/2$ and variance θ^2, which normalizes $E[K_i] = 1$. I estimate the dispersion parameter θ from the standard deviation of log intangible capital using the Compustat universe of firms during the sample period, including firm-years in which no first FDI project is observed. Because each potential project is associated with a parent firm at a given point in time, I use parent-year observations. The estimated dispersion is 2.049, and I set $\theta = 2$. I apply the same normalization to the data by dividing firm-year intangible capital by its empirical mean in the Compustat sample, so that the average firm-year has $K = 1$. Appendix Section D.1 provides diagnostics for the log-normal specification.

US and Destination Productivity. I measure destination productivity z using labor productivity per hour worked from the Penn World Table.[46] I normalize US productivity Z to one and define z as the average productivity of destination relative to the US over the sample period. Because local project productivity is $z = \tilde{z}^{1/\alpha}$, I set z equal to the US-normalized labor-productivity ratio raised to $1/\alpha$. The resulting productivity values z are 0.557 for all destinations, 0.798 for advanced economies, and 0.171 for non-advanced economies.

Labor Endowment per Foreign Project. I construct the host-country labor endowment per foreign project, L/M. I measure L using the average number of persons engaged over the sample period, obtained from the Penn World Table. Labor is expressed in thousands of workers. The mass of foreign projects M includes projects that may not undertake FDI and is therefore not directly observed. I proxy M by scaling up the cumulative number of observed US FDI projects $\tilde{M}$: $M = \nu_M \tilde{M}$. The scaling factor ν_M is the inverse of the sample-period FDI participation rate among Compustat firms.[47] This adjustment expands realized FDI projects to account for foreign projects associated with firms not observed undertaking FDI. For country groups, I aggregate labor and potential project masses before taking the ratio. The resulting values L/M are 60.254 for all destinations, 12.136 for advanced economies, and 134.590 for non-advanced economies. The larger value for non-advanced economies reflects the lower density of observed US FDI projects relative to the labor force in these destinations.

[45] This choice is motivated by Head et al. (2014) and Kondo et al. (2023), who show that log-normal heterogeneity provides a flexible alternative to the Pareto assumption and fits the firm-size distribution more broadly, rather than approximating only the upper tail. A quantile-quantile regression following these papers gives $\theta = 2.046$, nearly identical to the baseline estimate.

[46] The data are accessed through Our World in Data (Feenstra et al. - Penn World Table, 2025), and the original Penn World Table reference is Feenstra et al. (2015).

[47] The Compustat sample contains 7,404 unique firms, 2,471 of which are observed undertaking at least one FDI project. Thus, $\nu_M = 7,404/2,471$.

Table 4: Calibration Parameters

Parameter	Definition	All	Advanced	Non-advanced
Externally assigned parameters				
z	Productivity level of local projects	0.557	0.798	0.171
L/M	Labor per foreign project	60.254	12.136	134.590
ν_N	Effective-target share of local producers (%)	0.083	0.483	0.018
Internally calibrated parameters				
k	Local intangible capital	0.196	0.186	0.099
F	Fixed cost of GFDI	0.080	0.058	0.361
ψ	Search cost	0.021	0.009	0.107
c	Local project entry cost	0.050	0.040	0.058
ξ_1	Mean of integration-quality distribution $H(\eta)$	0.954	0.968	0.899

Notes: $H(\eta)$ is the integration-quality distribution, such that $\eta \sim \text{Beta}(\xi_1\xi_2, (1-\xi_1)\xi_2)$, with $\xi_2 = 100$.

Effective Acquisition Targets. In the model, the mass of active local producers N is determined endogenously, but only a subset of these producers are plausible acquisition targets for foreign projects. I therefore assume the effective acquisition-target pool is $N^T = \nu_N N$, where ν_N is the share of local producers that are acquisition-relevant and searchable by foreign acquirers.[48] Thus, ν_N is a calibrated parameter that maps the endogenous mass of local producers N into the effective target pool used in the matching function. I calibrate ν_N using the data counterpart, $\nu_N = N^{T,D}/N^D$, where $N^{T,D}$ is the empirical mass of acquisition targets and N^D is the empirical mass of local producers. I construct $N^{T,D}$ as observed annual M&A flows divided by 0.037, the annual probability that an effective target is acquired (David, 2021). I measure N^D using the number of enterprises with more than 50 employees.

Other Externally Assigned Parameters. The elasticity of substitution across differentiated varieties is $\sigma = 4$, following Arkolakis et al. (2018). This value implies a markup of 33 percent over marginal cost. I set the expenditure share on differentiated varieties to $\gamma = 0.51$, computed from OECD annual household final consumption expenditure data as the average goods share of total household consumption.[49] The labor share in the Cobb-Douglas production function is $\beta = 0.7$, and the local project's bargaining share is $\chi = 0.5$.

[48] The distinction between the total mass of local producers N and the effective target pool N^T is not merely a normalization. The full mass N clears the host labor and goods markets, while only the much smaller subset N^T enters the merger market. Setting $\nu_N = 1$, so that every local producer is an acquisition target, leaves the model without an equilibrium—an economy whose entire local sector is as thin as the acquisition-target pool cannot clear its factor and product markets. The parameter ν_N thus captures a substantive economic feature, the scarcity of acquirable targets relative to the broad producer base.

[49] The resulting value is close to the value 0.52 used by Irarrazabal et al. (2013), who calibrate the analogous expenditure-share parameter using Norwegian data.

Table 5: Target Moments and Model Fit

Targeted Moments	All Data / Model	Advanced Data / Model	Non-advanced Data / Model
Local producers relative to labor, N/L	4.674 / 4.674	5.363 / 5.485	4.578 / 3.793
Share of M&A, $\Pr(M\&A \mid FDI)$	0.416 / 0.416	0.557 / 0.545	0.197 / 0.148
Mean log K conditional FDI, $E[\log K \mid FDI]$	-0.310 / -0.310	-0.442 / -0.388	-0.071 / -0.541
Median GFDI–M&A log K gap, $\text{med}[\log K \mid GFDI] - \text{med}[\log K \mid M\&A]$	0.804 / 0.804	0.656 / 0.774	0.743 / 1.323
Acquirer-target profitability ratio, $ZE[K \mid \text{M\&A}]/(zk)$	4.953 / 4.953	4.953 / 4.812	4.953 / 8.565

Notes: This table presents target moments and model fit.

The latter choice is motivated by David (2021), who estimates an M&A bargaining parameter and finds that it is statistically indistinguishable from equal surplus splitting. Finally, I set the curvature parameter of the matching function to $\rho = 1.84$, following the estimate in Coşar et al. (2016), who use the same bounded matching function. Since their estimate comes from a labor-market application rather than an M&A market, I interpret this value as a functional-form benchmark.

5.2.2 Internally Calibrated Parameters and Target Moments

The internally calibrated objects are $\Theta \equiv (k, F, \psi, c, \xi_1)$, where k is local intangible capital, F is the fixed cost of GFDI, ψ is the search cost in the merger market, c is the local project entry cost, and ξ_1 governs the distribution of post-match integration quality, $H(\eta)$. I restrict the distribution $H(\eta)$ to a one-parameter beta family: $\eta \sim \text{Beta}\,(\xi_1\xi_2, (1-\xi_1)\xi_2)$, where $\xi_1 \in (0,1)$ is the mean of the distribution and ξ_2 controls its concentration. The baseline calibration sets $\xi_2 = 100$.[50] The mean value of η equals ξ_1. Since integration quality is not directly observed, I interpret ξ_1 as the calibrated mean of a latent post-match integration-quality distribution.

Given a candidate Θ, I solve the model equilibrium characterized by (K_m, K_g, n, N, B) and compute the corresponding model-implied moments. I then choose Θ to minimize the distance between model and data moments. Although the calibrated objects are jointly determined by all targeted moments through equilibrium, each target is chosen to capture a main margin of the model. The number of local producers relative to labor N/L disciplines the entry cost of local projects c. The acquirer-target profitability ratio, $ZE[K \mid M\&A]/(zk)$,

[50]This value avoids the endpoint spike that can arise under low-concentration beta distributions when the estimated location parameter is close to the upper support.

Table 6: Equilibrium Objects

Object	Definition	All	Advanced	Non-advanced
Equilibrium objects				
K_m	Lower search cutoff	0.142	0.102	0.099
K_g	Upper search cutoff	1.731	4.119	0.250
n	Mass of searching foreign projects	0.388	0.513	0.184
N	Mass of local producers	281.636	66.570	510.466
B	Aggregate demand shifter	0.461	0.271	3.064
Model-implied objects				
$\tilde{K}$	GFDI zero profit threshold	0.174	0.213	0.118
K_η	Integration threshold	0.261	0.250	0.210
$\nu_N N$	Acquisition-target mass	0.234	0.321	0.092
μ	Matching probability	0.503	0.517	0.438
Bz	Productivity-adjusted demand shifter	0.257	0.216	0.524

Notes: The computation normalizes the mass of foreign projects to $M = 1$, so n, N, and $\nu_N N$ are measured relative to M. The cutoffs K_m and K_g are expressed in normalized K-units. The cutoff magnitudes can be interpreted as multiples of average intangible capital or as percentiles of $G(K)$. The shifter B is an endogenous equilibrium object and is not a directly observed demand measure.

anchors the level of local intangible capital k. I target a log acquirer-target productivity gap of 1.6, following David (2021). The share of M&A conditional on FDI, $\Pr(M\&A \mid FDI)$, reflects the relative attractiveness of acquisition and search, which is shaped by the search cost ψ, the fixed cost of GFDI F, and the integration-quality distribution $H(\eta)$. The remaining two moments use parent intangible capital to capture selection and sorting over the underlying distribution $G(K)$. The mean log intangible capital of FDI projects, $E[\log K \mid FDI]$, targets overall selection into FDI relative to no FDI. The median gap in log intangible capital between GFDI and M&A projects, $\text{med}[\log K|GFDI] - \text{med}[\log K|M\&A]$, disciplines sorting across entry modes.[51]

5.2.3 Model Fit and Interpretation

Table 5 reports the target moments and the model fit. The all-destination calibration matches the five targeted moments very closely. The calibrated parameters are reported in Table 4. The mean of the integration-quality distribution, $E[\eta] = \xi_1$, is 0.954. This value implies that realized M&A matches are selected toward high integration compatibility, and the integration-loss mechanism affects the expected merger surplus and the location of the

[51] The median gap captures the relative location of GFDI and M&A projects in $G(K)$, while being less sensitive to extreme observations and the skewness in the level distribution of K.

upper search cutoff K_g. Table 6 reports the equilibrium objects implied by the calibrated parameters. In the all-destination calibration, the lower search cutoff is $K_m = 0.142$, which indicates that projects above the 51st percentile of the distribution $G(K)$ invest. The upper search cutoff is $K_g = 1.731$, corresponding to approximately the 90th percentile of $G(K)$. Hence, the top 10 percent of foreign projects choose GFDI without search. The mass of projects in the search region is $n = 0.388$, which means that 38.8 percent of foreign projects search for local firms. The model-implied matching probability is $\mu = 0.503$, so roughly half of the searching projects are matched with local targets.

The internally calibrated parameters highlight the contrast between advanced and non-advanced economies. The advanced-economy calibration has higher local intangible capital k and a higher mean integration quality $E[\eta]$ than the non-advanced-economy calibration. By contrast, non-advanced economies require a higher GFDI fixed cost F, a higher search cost ψ, and a higher local entry cost c. These parameter values are consistent with a more difficult environment for both local entry and M&A implementation. The equilibrium mass of searching foreign projects n is also smaller in non-advanced economies than in advanced economies. Although advanced economies have a thicker acquisition-target pool, $\nu_N N$, their matching probability μ is lower because the mass of searching foreign projects is much larger, creating more congestion in the search market. The demand shifter B is an endogenous object determined within each destination-sample equilibrium. In particular, low local productivity z and low local intangible capital k require a larger B to support local entry in non-advanced economies.

The advanced-economy calibration fits the targeted moments reasonably well. In contrast, the non-advanced-economy calibration reveals a sharper model-data tension. The model generates a low M&A share in non-advanced economies partly by lowering the upper search cutoff K_g, which expands the no-search GFDI region. As a result, the model underpredicts the mean log intangible capital K of FDI projects and overpredicts the GFDI–M&A sorting gap. This suggests that the parsimonious target-side structure, with homogeneous local intangible capital and a single search friction, captures the qualitative mechanism but does not fully reproduce the observed sorting of K across entry modes in non-advanced economies.

5.3 Welfare and Source of Inefficiency

The decentralized equilibrium need not be efficient. Foreign projects choose search effort and entry mode based on private payoffs, whereas the host country evaluates the allocation through household income and the price index for differentiated goods. I analyze welfare from the perspective of the representative household in the host country.

5.3.1 Host Country Welfare

Maximizing the Cobb-Douglas preferences in equation (2) subject to the budget constraint $y_0 + PC = I$ yields indirect utility $\mathcal{U} = \log I - \gamma \log P$ up to an additive constant. Here, I denotes host-country household income, P is the CES price index and γ is the expenditure share on differentiated varieties. I define host-country welfare as $W = I/P^{\gamma}$. Using the local free-entry condition, host-country income reduces to wage income, $I = L$, as shown in equation (22). Therefore,

$$\log W = \log L - \gamma \log P. \tag{24}$$

Because the labor endowment L is fixed, changes in host-country welfare operate only through the differentiated-good price index P.

Next, consider the aggregate expenditure condition in equation (23). Suppressing the dependence of $\bar{\eta}$ on B for notational simplicity, define the bracketed object in that equation as the effective-input aggregate, $\mathbb{Y}$:

$$\mathbb{Y} = \underbrace{M \int_{K_g}^{\infty} ZK\, dG(K)}_{\text{no-search GFDI}} + \underbrace{(1-\mu)\, M \int_{\tilde{K}}^{K_g} ZK\, dG(K)}_{\text{search, no match}\rightarrow\text{GFDI}} + \underbrace{\mu M \int_{K_\eta}^{K_g} H(\bar{\eta})\, ZK\, dG(K)}_{\text{match, low }\eta\rightarrow\text{GFDI}}$$

$$+ \underbrace{\mu M \int_{K_m}^{K_g} \int_{\bar{\eta}}^{1} Z(k + \eta K)\, dH(\eta)\, dG(K)}_{\text{match, high }\eta\rightarrow\text{M\&A}} + \underbrace{\left(N - \mu M \int_{K_m}^{K_g} \left[1 - H(\bar{\eta})\right] dG(K) \right) zk}_{\text{standalone locals}}.$$

Then equation (23) can be written as $\gamma L = \frac{\sigma}{\alpha(\sigma-1)} B\mathbb{Y}$. Since γL is fixed, B is inversely proportional to $\mathbb{Y}$. Using the definitions $A \equiv \gamma I P^{\sigma-1}$ and $B \equiv \frac{\alpha}{\beta}\left[\frac{\beta(\sigma-1)}{\sigma} A^{1/\sigma}\right]^{\frac{\sigma}{\alpha(\sigma-1)}}$, $P \propto \mathbb{Y}^{-\alpha}$. Thus, host-country welfare can be summarized by

$$\log W = \gamma\alpha \log \mathbb{Y} + \Xi,$$

where Ξ is a constant. Maximizing host-country welfare is therefore equivalent to maximizing $\mathbb{Y}$.[52]

[52] The welfare elasticity $\gamma\alpha$ indicates that the uniform CES mark-up is not a distortion because it is common across all varieties and does not distort relative prices or the relative attractiveness of M&A versus GFDI. The markup scales P without altering the composition of $\mathbb{Y}$.

To characterize the welfare effect of changing the upper search cutoff, fix $\{n, N, B\}$ and differentiate $\mathbb{Y}$ with respect to K_g. Applying the Leibniz rule gives

$$\left.\frac{\partial \mathbb{Y}}{\partial K_g}\right|_{n,N,B} = \mu M g(K_g) \int_{\bar{\eta}(K_g)}^{1} \Big[Z(k + \eta K_g) - Z K_g - z k \Big] \, dH(\eta). \tag{25}$$

The bracketed term is the change in the effective-input aggregate when the marginal project at the upper search cutoff is implemented through M&A rather than GFDI. Specifically, M&A replaces one foreign greenfield variety, ZK_g, and one stand-alone local variety, zk, with a merged variety, $Z(k + \eta K_g)$. This marginal change is weighted by the mass of marginal projects $Mg(K_g)$, the matching probability μ, and the distribution of integration-quality draws $H(\eta)$. This observation leads to the following proposition.

Proposition 2 *For a given lower search cutoff K_m, and holding $\{n, N, B\}$ fixed, define the expected marginal contribution of the upper search cutoff K_g to the effective-input aggregate $\mathbb{Y}$, $\Gamma(K_g) \equiv \int_{\bar{\eta}(K_g)}^{1} \Big[Z(k + \eta K_g) - Z K_g - z k \Big] dH(\eta)$. Let h denote the continuous density of H, and define its hazard rate as $R_H(\eta) \equiv h(\eta)/[1 - H(\eta)]$. Suppose the integration-quality distribution satisfies the regularity condition, $B(Z-z)k/F > (1-\bar{\eta}(K_g))R_H(\bar{\eta}(K_g))$, at every zero of Γ. Then Γ crosses zero exactly once and from above. Hence, there exists a unique interior cutoff $\hat{K}_g$ that maximizes $\mathbb{Y}$, defined by $\Gamma(\hat{K}_g) = 0$. When the economy receives more M&A ($K_g > \hat{K}_g$) or less M&A ($K_g < \hat{K}_g$) than the optimal cutoff, host welfare falls below its maximum.*

Proof. See Appendix A.6.

In the calibrated economies, I verify numerically that $\Gamma(K_g)$ has a unique zero over the admissible range and that the local regularity condition holds at this zero.

5.3.2 Source of Inefficiency

Proposition 2 characterizes the upper search cutoff $\hat{K}_g$ that maximizes host-country welfare. The decentralized economy generally does not select this cutoff because foreign projects choose search and entry mode to maximize their own repatriated payoffs rather than the host-country effective-input aggregate $\mathbb{Y}$. To see the source of the wedge, compare the host-country optimality condition with the decentralized upper-cutoff condition, holding $\{n, N, B\}$ fixed.

The host-country optimal cutoff satisfies

$$\Gamma(\hat{K}_g) = \int_{\bar{\eta}(\hat{K}_g)}^{1} \Big[Z(k + \eta \hat{K}_g) - Z\hat{K}_g - zk \Big] dH(\eta) = 0.$$

By contrast, the decentralized upper search cutoff is determined by equation (17). Using the expression for merger surplus Σ_2 in equation (8), the private search condition can be rearranged as

$$\Gamma(K_g) = \int_{\bar{\eta}(K_g)}^{1} \Big[Z(k + \eta K_g) - ZK_g - zk \Big] dH(\eta) = \frac{1}{B} \left[\frac{\psi}{\mu(1-\chi)} - F\left(1 - H(\bar{\eta}(K_g))\right) \right]. \quad (26)$$

The host-country condition sets $\Gamma(\hat{K}_g)$ equal to zero, whereas the decentralized condition sets $\Gamma(K_g)$ equal to the wedge on the right-hand side of equation (26). Because $\Gamma(K_g)$ crosses zero once and from above, a positive wedge implies $K_g < \hat{K}_g$, so the decentralized economy generates too little M&A from the host-country perspective. A negative wedge implies $K_g > \hat{K}_g$, so the decentralized economy generates too much M&A. The wedge contains four externalities.

Externalities Standard to Search and Matching. First, the factor $1/(1-\chi)$ reflects an *appropriability problem*. A foreign project receives only the share $1-\chi$ of its merger surplus, whereas the host country evaluates the full gains from mergers. This externality leads to fewer searching foreign projects and pushes the decentralized equilibrium toward too little M&A. Second, the matching probability μ creates a *congestion externality*. In the cutoff comparison above, the mass of searchers n is fixed. If n is allowed to adjust, however, an additional searching project reduces the matching probability faced by other searchers, $\partial\mu/\partial n < 0$. Individual foreign projects take μ as given, while a planner internalizes the effect of additional search and cares about the marginal change in the probability of matching $\partial\mu/\partial n$. This congestion force pushes the decentralized economy toward excessive search and, therefore, too much M&A. Appendix A.7 derives this term explicitly.

Externalities Specific to Host Welfare. The remaining two forces arise because host-country welfare depends on $\mathbb{Y}$, rather than on foreign projects' private search-and-entry surplus. First, the search cost ψ enters the private condition but not the host-country welfare condition. This is because ψ uses source-country resources and does not enter host-country income. The host country therefore attaches no cost to an additional searcher and counts more search than the decentralized equilibrium. This externality pushes the decentralized equilibrium toward too little search and too little M&A. Second, the term $F(1-H(\bar{\eta}))$ enters the decentralized condition with a negative sign. When a searching project chooses M&A over GFDI, it avoids paying the fixed cost of GFDI F. This saving raises the private value of M&A relative to GFDI. However, F does not enter the host-country welfare; it affects only residual employment and leaves $I = L$ unchanged. The decentralized economy therefore places too much weight on the fixed-cost saving from M&A, which pushes toward too much M&A.

Taken together, the sign of the wedge determines the direction in which a welfare-improving policy moves the allocation, and this direction corresponds to the GFDI–M&A trade-off. The function $\Gamma(K_g)$ measures the host country's marginal gain from converting the project at K_g from GFDI to M&A. A positive wedge, which implies $K_g < \hat{K}_g$, indicates the host would gain from shifting marginal projects toward M&A. This marginal shift replaces a GFDI variety ZK_g and a stand-alone local variety zk with a merged variety $Z(k+\eta K_g)$, which raises the effective-input aggregate $\mathbb{Y}$. Conversely, a negative wedge implies $K_g > \hat{K}_g$, so the host country would gain from shifting marginal projects toward GFDI. A welfare-improving policy therefore raises welfare by reallocating foreign projects across entry modes—toward M&A when the wedge is positive, and toward GFDI when it is negative.

5.3.3 The Inefficiency Wedge in the Calibrated Economies

Equation (26) shows that the decentralized upper-cutoff condition differs from the host-country optimality condition by the inefficiency wedge

$$B\Gamma(K_g) = \underbrace{\psi/[\mu(1-\chi)]}_{\text{search-cost component}} - \underbrace{F\left[1 - H(\bar{\eta}(K_g))\right]}_{\text{fixed-cost-saving component}} . \tag{27}$$

Since $B > 0$, the wedge has the same sign as $\Gamma(K_g)$. Table 7 reports the two components of the wedge and the implied direction of inefficiency for each calibration. The all-country calibration lies close to the efficient margin: the search-cost component is almost exactly offset by the fixed-cost-saving component.

The advanced- and non-advanced-economy calibrations lie on opposite sides of the efficient margin. In the advanced-economy calibration, the search cost is small ($\psi = 0.009$), so the search-cost component is weak. The fixed-cost-saving component slightly dominates, which generates a small negative wedge. The wedge implies a marginal excess of M&A relative to the host-welfare optimum, although the advanced-economy calibration remains close to the efficient composition. By contrast, in the non-advanced-economy calibration, the search cost is high ($\psi = 0.107$). The force pushing toward too little M&A, $\psi/[\mu(1-\chi)]$,

Table 7: Inefficiency Wedge by Calibration

	$\psi/[\mu(1-\chi)]$	$F[1-H(\bar{\eta})]$	Wedge, $B\Gamma$	Prediction
All	0.084	0.080	0.004	$\approx$ Efficient
Advanced	0.035	0.054	-0.019	$\approx$ Efficient (slight excess M&A)
Non-advanced	0.487	0.361	0.126	Too little M&A / too much GFDI

Notes: The table reports the components of the inefficiency wedge in equation (26), evaluated using the calibrated parameters. By Proposition 2, a positive wedge implies $K_g < \hat{K}_g$, so the decentralized economy generates too little M&A relative to the host-welfare optimum. A negative wedge implies $K_g > \hat{K}_g$, so the decentralized economy generates too much M&A.

dominates the private fixed-cost saving from choosing M&A over GFDI, $F[1-H(\bar{\eta})]$. The wedge is therefore positive: the host country receives too little M&A, and policies that encourage M&A can raise welfare.

Overall, the inefficiency is concentrated in the non-advanced-economy calibration, whereas the advanced-economy calibration remains close to the efficient margin. Appendix D.3 decomposes the difference in the inefficiency wedge between the advanced- and non-advanced-economy calibrations by cumulatively replacing non-advanced-economy parameters with their advanced-economy values. The decomposition shows that high search cost ψ in the non-advanced-economy calibration is a key force behind the positive inefficiency wedge, which implies too little M&A. I therefore focus the counterfactual policy analysis in the next subsection on the non-advanced-economy calibration.

5.4 Counterfactual Analysis

I now introduce policy instruments that act on the inefficiency wedge and quantify their welfare consequences in non-advanced economies.

5.4.1 Balanced Budget Policy in Non-advanced Economies

I consider two fiscal instruments for the non-advanced economy calibration: a subsidy $s_m \geq 0$ paid to the foreign acquirer for each completed cross-border M&A, and a tax $\tau_g \geq 0$ levied on each greenfield project.[53] Both instruments are in units of the homogeneous good. The instruments enter project-level payoffs and propagate through the equilibrium conditions.[54]

[53]The subsidy is paid to the foreign acquirer and is not part of the bargaining surplus. This setting is consistent with how investment promotion incentives are structured in practice—tax holidays, investment credits, and cash grants often accrue to the foreign investor rather than to the acquisition bargain.

[54]The payoff and cutoff conditions affected by the subsidy and tax are presented in Appendix D.4.

With the policy in place, the subsidy and tax enter the search-cutoff conditions in equations (16) and (17). A searching acquirer receives the subsidy s_m on every match that completes as M&A. The lower cutoff K_m governs projects whose outside option is no investment and which complete M&A for any integration draw if matched. Conditional on matching, the subsidy is collected with probability one:

$$\mu\Big[(1-\chi)\int_0^1 \Sigma_1(K_m,\eta)\,dH + s_m\Big] = \psi,$$

where $\Sigma_1(K_i,\eta_i) \equiv B[(Z-z)k + Z\eta_i K_i]$ for $K_i \in (0,\tilde{K})$. The upper cutoff K_g governs projects whose outside option is greenfield. A matched project completes M&A only when $\eta \geq \bar{\eta}(K_g)$, so the subsidy is collected with probability $1 - H(\bar{\eta}(K_g))$:

$$\mu\Big[(1-\chi)\int_{\bar{\eta}(K_g)}^1 \Sigma_2^\tau(K_g,\eta)\,dH + s_m\big(1 - H(\bar{\eta}(K_g))\big)\Big] = \psi,$$

where $\Sigma_2^\tau(K_i,\eta_i) \equiv B[(Z-z)k - Z(1-\eta_i)K_i] + F + \tau_g$ for $K_i \in [\tilde{K},\infty)$. For projects with GFDI as the outside option, the GFDI tax τ_g raises the value of M&A relative to GFDI Σ_2^τ by increasing the cost of greenfield entry.

In terms of Figure 1, the policy instruments shift the relevant expected-gain-from-search lines upward. The M&A subsidy adds to the acquirer's expected payoff from completed mergers, while the GFDI tax raises the merger surplus at the upper cutoff by weakening the GFDI outside option. Holding the aggregate objects fixed, the M&A subsidy expands the search region by lowering K_m and raising K_g.

I consider a balanced-budget policy in which the host country finances the M&A subsidy with the GFDI tax. Net policy revenue is tax revenue less subsidy payments:

$$T = \tau_g \Phi_g M - s_m \Phi_m M,$$

where Φ_g and Φ_m are GFDI and M&A shares among potential foreign projects. Net policy revenue T enters host-country income as $I = L + T$. Under a balanced budget policy, $\tau_g \Phi_g = s_m \Phi_m$, so net policy revenue is zero, $T = 0$, and $I = L$. For a given subsidy s_m, the GFDI tax τ_g is determined endogenously by this balanced-budget condition.

Using the welfare representation (24), the welfare change is

$$\Delta \log W = \Delta \log I - \gamma \Delta \log P = -\gamma \Delta \log P.$$

Table 8: Counterfactual Analysis: Non-advanced Economies

Policy	s_m	τ_g	s_m/ψ	τ_g/F	$\Delta \log W$ (%)	μ	$Pr(M\&A\|FDI)$
Laissez-faire	0	0	0	0	0	0.438	0.148
Moderate subsidy	0.10	0.020	0.93	0.05	0.0027	0.305	0.165
Optimal	0.45	0.097	4.21	0.27	0.0040	0.177	0.177
High subsidy	0.55	0.120	5.14	0.33	0.0039	0.159	0.179

Notes: The table reports balanced-budget policies in the non-advanced-economy calibration. Each row pairs an M&A subsidy s_m with the GFDI tax τ_g that finances it. The columns s_m/ψ and τ_g/F express the instruments relative to the search cost and the GFDI fixed cost, respectively. $\Delta \log W$ is the welfare change relative to laissez-faire, in percent. μ is the matching probability and $\Pr(M\&A|FDI)$ is the model-implied M&A share among FDI projects.

Under the balanced budget, host income is unchanged, $\Delta \log I = 0$, so the entire welfare effect operates through the price index.

5.4.2 The Optimal Subsidy and the Role of Target Scarcity

Table 8 reports the balanced-budget policies. Host welfare is maximized at an optimal subsidy, $s_m^\star = 0.45$: welfare rises as the subsidy increases from zero, peaks at $s_m^\star$ with $\Delta \log W = 0.004\%$, and declines thereafter. As the subsidy draws additional foreign projects into the merger market, the matching probability falls steadily, from $\mu = 0.438$ at laissez-faire to $\mu = 0.177$ at the optimum. Beyond $s_m^\star$, the congestion cost imposed on marginal searchers outweighs the composition gain from additional M&A, so welfare declines. The optimal subsidy is large relative to the search cost, $s_m^\star/\psi = 4.21$, while the GFDI tax that finances it remains moderate, $\tau_g^\star/F = 0.27$.

The welfare gain at the optimum is positive but small, even though the laissez-faire inefficiency wedge is large, $B\Gamma = 0.126$ (Table 7). These two facts are reconciled by the distinction between the *marginal* distortion and the *realized* quantity response. To first order, the welfare gain depends on the product of the wedge and the change in composition that the policy induces. A large wedge therefore delivers a large gain only if the policy moves composition substantially. In this calibration, it does not: the optimal policy raises the M&A share only from 0.148 to 0.177 because the merger market is target-constrained. With a thin pool of acquirable local firms, additional searchers are absorbed mainly by a lower matching probability rather than by completed mergers, so the subsidy partly crowds itself out through congestion.

This finding is the counterpart of the structural role of ν_N in the calibration. The model distinguishes the total mass of local producers N, which clears the labor and goods markets, from the effective pool of acquisition targets $N^T = \nu_N N$, which enters the merger market.[55] The share ν_N is small in the non-advanced-economy calibration: only a limited fraction of local firms are realistic cross-border acquisition targets. As a result, the expected acquisition premium in the local entry decision is small. The M&A subsidy, which operates primarily through the acquisition margin, does little to induce additional local entry. This is why the optimal subsidy, though large, moves the M&A share only modestly and delivers only a small welfare gain.

The substantive message is therefore not that the inefficiency is small. Rather, the host country is far from the efficient allocation, but the welfare improvement from M&A promotion is bounded by the thinness of the merger market. Policies that enlarge the pool of acquirable firms, rather than subsidize transactions within a fixed thin pool, lie outside the present model but are the natural implication of this result.

6 Conclusion

This paper studies how multinational firms choose between greenfield foreign direct investment (GFDI) and cross-border mergers and acquisitions (M&A), and how this entry-mode choice affects welfare in host countries. I develop a general equilibrium search model in which foreign projects differ in intangible capital and M&A requires matching with an acquirable local target. The model generates selection across FDI modes. Projects with lower intangible capital benefit more from acquiring local intangible assets and are therefore more likely to choose M&A, whereas projects with higher intangible capital rely more on their own assets and are more likely to choose GFDI.

I test these predictions using a novel US project-level dataset that links GFDI projects, cross-border M&A deals, and parent-firm financial information. The empirical results support the model mechanism. Parent firms with lower internally accumulated intangible capital are more likely to enter foreign markets through M&A, while more intangible-intensive parent firms are more likely to use GFDI. I also find that M&A is more likely in host markets with thicker pools of potential targets and richer local intangible assets.

The paper then uses the model to study host-country welfare. The decentralized allocation need not be efficient because foreign projects make search and entry-mode decisions based on private payoffs, while the host country values the resulting composition of foreign and local varieties. The calibrated model implies that the direction of the inefficiency differs

[55] See footnote 48 for the empirical motivation and construction of ν_N.

across destination groups. Advanced economies lie close to the efficient margin, whereas the non-advanced-economy calibration features too little M&A relative to the host-country optimum. This result motivates a counterfactual policy that subsidizes completed M&A and finances the subsidy with a tax on GFDI.

In the non-advanced-economy calibration, the balanced-budget policy raises host welfare, but the gain is small. The reason is target scarcity. Only a thin subset of local producers is effectively acquirable, so additional search lowers the matching probability rather than translating one-for-one into completed mergers. Thus, the policy implication is not simply that host countries should subsidize M&A. M&A promotion is more effective when the merger market has enough acquirable targets.

The analysis of host-country welfare points to two policy implications. First, the high cost of searching for local targets is a key driver of inefficiency in non-advanced economies, where too little M&A occurs relative to the host-country optimum. Policies that lower this search cost—by helping foreign acquirers identify and evaluate potential local targets—address the friction that generates the inefficiency directly. Second, the welfare gain from promoting M&A is limited by the availability of acquirable firms. When the target pool is thin, transaction subsidies are dissipated through congestion rather than translated into completed mergers. Policies that expand the set of acquisition-ready local firms—for example, by improving accounting transparency, legal compliance, managerial systems, or corporate governance—would relax the target scarcity that limits the effect of transaction subsidies. I leave the evaluation of these policies to future work.

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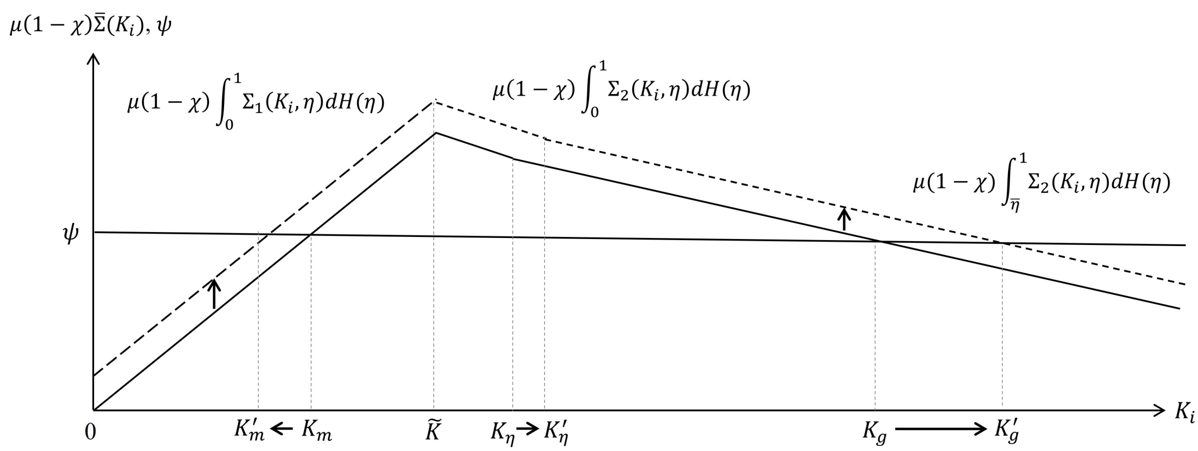
Appendix A Model Appendix

A.1 Discussion: Assumption of Homogeneous Local Intangibles

In the main model, I assume that local projects' intangible capital, k, is homogeneous and fixed. This assumption is motivated by data limitations: even with access to Orbis and Compustat Global, information on the intangible capital of target projects is rarely available, particularly for small firms involved in cross-border M&A. While this assumption simplifies the analysis, some readers may be concerned that abstracting from matching dynamics—such as sorting or complementarities between acquirer and target projects—could distort the model's implications.

To address this concern, I examine how foreign projects' investment decisions respond to changes in local intangible capital. Figure A.1 shows that when the level of local intangible capital k increases, the merger surplus curves shift upward. As a result, the threshold K_m (below which projects do not invest) falls, while K_g (above which projects choose GFDI without searching) rises. As local intangibles increase, the gain from merging becomes larger, and a greater share of foreign projects find it optimal to invest via M&A.

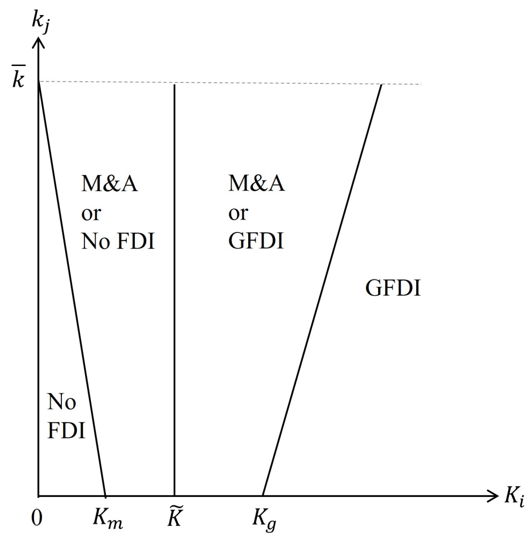
Figure A.1: Effect of an Increase in Local Intangible Capital on Investment Thresholds



Note: An increase in local intangible capital k shifts the expected gain from search upward, which lowers the no-FDI threshold K_m and raises the thresholds, K_η and K_g. Consequently, the search region $[K_m, K_g)$ and the range of foreign projects that undertake M&A upon matching expand.

Figure A.2 shows that the partition of foreign projects into investment modes—no FDI, M&A, and GFDI—remains unchanged even when local intangible capital varies. That is, the thresholds K_m, $\tilde{K}$, and K_g that govern project-level investment decisions continue to hold even when local intangible capital k is heterogeneous. The simplifying assumption of homogeneous local intangibles does not affect the sorting patterns emphasized in Proposition 1.

Figure A.2: Investment Mode Partition under Heterogeneous Local Intangibles



Note: The figure shows the cutoff thresholds K_m, $\tilde{K}$, and K_g that define the search and investment behavior of foreign projects. These thresholds remain valid when local intangible capital k is allowed to vary across local projects. For visual simplicity, the intermediate threshold K_η is omitted because it does not change the set of possible investment modes in that region.

A.2 Anecdotal Evidence on Intangible-Asset Integration in M&A

In the model, the production function of an M&A project is

$$y_i^m = \tilde{Z}(k + \eta_i K_i)^\alpha \ell_i^\beta,$$

where the acquiring foreign project uses the target project's intangible capital k and combines it with the effective portion $\eta_i K_i$, $\eta_i \in (0,1)$, of its own intangible capital. This appendix provides anecdotal examples illustrating two features of this formulation: target intangible assets can remain valuable after acquisition, while the acquirer's intangible assets may be only partially integrated or redeployed after acquisition.

Renault–Nissan Alliance. The Renault–Nissan Alliance illustrates both the use of local intangible assets and the limits to integrating foreign intangible assets. Renault vehicles were sold through Nissan's local sales network in Japan: Nissan dealers began selling Renault vehicles in May 2000, and 71 exclusive or dual-brand dealer outlets were operating by the end of 2002 (Renault-Nissan Alliance, 2003). This illustrates how a foreign firm's brand and product assets can be combined with a local partner's sales network. At the same time, the integration of technical and organizational assets was partial and difficult. Segrestin (2005) documents that the development of the first joint Renault–Nissan platform involved uncertain synergies, fewer shared components than initially expected, and tensions over specifications and validation procedures. For example, Renault and Nissan ultimately developed fuel tanks separately after disagreements over validation criteria. This case illustrates why only the effective portion of the foreign project's intangible capital may enter the merged project.

AB InBev acquisitions. Alviarez et al. (2025) provide evidence from beer and spirits markets that is consistent with both the preservation of acquired local intangible assets and limited post-acquisition redeployment of brand assets. Their discussion of AB InBev's acquisition of Grupo Modelo illustrates limited post-acquisition redeployment of the acquirer's brand assets. AB InBev already sold Budweiser and Bud Light in Mexico before the acquisition, and after acquiring Grupo Modelo it brought only one additional AB InBev brand, Stella Artois, to Mexico. I use this case to illustrate that the acquirer's intangible capital may not be frictionlessly transferred to the acquired market.

The Dominican Republic case illustrates a different aspect of the model: the preservation and combination of local and foreign brand assets under common ownership. Before AB InBev acquired the beer brands owned by Grupo León Jimenes, Presidente and Presidente Light dominated the lager and light beer segments in the Dominican Republic, while AB InBev had smaller shares with Brahma and Brahma Light. The acquisition combined AB InBev's existing global brands with local star brands under common ownership. This example is consistent with the model's M&A production structure, in which the merged project combines the target project's intangible capital with only the effective portion of the foreign project's intangible capital, $k + \eta_i K_i$.

A.3 Share of GFDI Projects

Analogous to the M&A share in equation (12), the share of GFDI investment, Φ_g, can be expressed as:

$$\Phi_g = [1 - G(K_g)] + (1 - \mu(n, N))[G(K_g) - G(\tilde{K})] + \mu(n, N) \int_{K_\eta}^{K_g} H(\bar{\eta}(K))dG(K), \quad \text{(A.1)}$$

where the three terms represent: (i) GFDI projects that do not search ($K_i > K_g$); (ii) foreign projects that search but do not match; (iii) foreign projects that search and match but draw $\eta_i < \bar{\eta}(K_i)$ and thus choose GFDI over M&A.

A.4 Derivation of Expenditure Condition

Using CES demand (3), revenue for a differentiated variety ω is

$$r_\omega = p_\omega y_\omega = A^{1/\sigma} y_\omega^{\frac{\sigma-1}{\sigma}}.$$

For a variety with production function $y_\omega = \tilde{Z}_\omega K_\omega^\alpha \ell_\omega^\beta$, revenue becomes

$$r_\omega = A^{1/\sigma} (\tilde{Z}_\omega K_\omega^\alpha \ell_\omega^\beta)^{\frac{\sigma-1}{\sigma}}.$$

The firm chooses labor to maximize operating profit $r_\omega - \ell_\omega$ since the wage is normalized to one. The first-order condition is

$$\ell_\omega = \frac{\beta(\sigma - 1)}{\sigma} r_\omega. \quad \text{(A.2)}$$

Using this first-order condition and the normalization $\alpha = \frac{\sigma}{\sigma-1} - \beta$, optimal operating profit is

$$\hat{\pi}_\omega = r_\omega - \frac{\beta(\sigma - 1)}{\sigma} r_\omega = \frac{\alpha(\sigma - 1)}{\sigma} r_\omega,$$

which shows that revenue is proportional to optimized operating profit $r_\omega = \frac{\sigma}{\alpha(\sigma-1)} \hat{\pi}_\omega$.

The operating-profit components before fixed costs and acquisition payments are given by equation (4): BZK_i for GFDI projects, $BZ(k + \eta_i K_i)$ for M&A projects, and Bzk for non-acquired local projects. Therefore, aggregate revenue from active differentiated varieties is obtained by multiplying the corresponding operating-profit components by $\frac{\sigma}{\alpha(\sigma-1)}$.

Because households spend a fraction γ of income I on differentiated varieties, aggregate expenditure on differentiated varieties is γI. Expenditure clearing requires this amount to equal the total revenue of all active differentiated varieties in the host country. Hence,

$$\gamma I = \frac{\sigma}{\alpha(\sigma-1)} B \Bigg[M \int_{K_g}^{\infty} ZK\, dG(K) + [1-\mu(n,N)]\, M \int_{\widetilde{K}(B)}^{K_g} ZK\, dG(K) \\ + \mu(n,N) M \int_{K_\eta(B)}^{K_g} H(\bar{\eta}(K,B)) ZK\, dG(K) \\ + \mu(n,N) M \int_{K_m}^{K_g} \int_{\bar{\eta}(K,B)}^{1} Z(k+\eta K)\, dH(\eta) dG(K) \\ + \left[N - \mu(n,N) M \int_{K_m}^{K_g} [1 - H(\bar{\eta}(K,B))]\, dG(K) \right] zk \Bigg].$$

The right-hand side is the total revenue of all active differentiated varieties in the host country. The five terms inside the brackets correspond, respectively, to: (i) foreign projects with $K_i \in [K_g, \infty)$ that enter through GFDI without search; (ii) foreign projects with $K_i \in [\tilde{K}, K_g)$ that search but fail to match, and therefore enter through GFDI; (iii) foreign projects with $K_i \in [K_\eta, K_g)$ that match but draw low integration quality $\eta_i < \bar{\eta}$, and therefore choose GFDI; (iv) foreign projects with $K_i \in [K_m, K_g)$ that match and draw sufficiently high integration quality, $\eta_i \geq \bar{\eta}$ complete M&A; and (v) stand-alone local projects. Using $I = L$, this condition becomes equation (23).

A.5 Derivation of the Labor Demand for Differentiated Varieties

This appendix derives labor used in the variable production of active differentiated varieties. The first-order condition in equation (A.2) implies the labor demand for variety ω

$$\ell_\omega = \frac{\beta(\sigma-1)}{\sigma} r_\omega.$$

Aggregating over all active differentiated varieties gives

$$L^\Omega = \int_\Omega \ell_\omega d\omega = \frac{\beta(\sigma-1)}{\sigma} \int_\Omega r_\omega d\omega.$$

Using the expenditure condition (23), $\gamma L = \int_\omega r_\omega d\omega$, labor used in the variable production of active differentiated varieties is

$$L^\Omega = \frac{\beta(\sigma-1)}{\sigma} \gamma L.$$

A.6 Proof of Proposition 2

Let $\Delta(K,\eta) \equiv Z(k+\eta K) - ZK - zk = (Z-z)k - Z(1-\eta)K$, so that $\Gamma(K_g) = \int_{\bar{\eta}(K_g)}^{1} \Delta(K_g,\eta)\,dH(\eta)$. Throughout the proof, $\{n, N, B\}$ are fixed. Define $\Gamma(K_g) \equiv \int_{\bar{\eta}(K_g)}^{1} \Big[Z(k+\eta K_g) - ZK_g - zk\Big]\,dH(\eta)$. From equation (25),

$$\left.\frac{\partial \mathbb{Y}}{\partial K_g}\right|_{\mu,N,B} = \mu M g(K_g)\Gamma(K_g).$$

The sign of $\partial\mathbb{Y}/\partial K_g$ is the sign of $\Gamma(K_g)$. An interior cutoff K_g is a critical point of $\mathbb{Y}$ if and only if $\Gamma(K_g) = 0$.

Applying the Leibniz rule to $\Gamma(K_g)$ gives

$$\Gamma'(K_g) = \int_{\bar{\eta}(K_g)}^{1} \frac{\partial \Delta(K_g,\eta)}{\partial K_g}\,dH(\eta) - \Delta(K_g,\bar{\eta}(K_g))h(\bar{\eta}(K_g))\bar{\eta}'(K_g). \tag{A.3}$$

For the relevant region $K_g \geq K_\eta$, the post-match integration cutoff is

$$\bar{\eta}(K_g) = 1 - \frac{(Z-z)k + F/B}{ZK_g}.$$

Thus, the derivative of $\bar{\eta}(K_g)$ is

$$\bar{\eta}'(K_g) = \frac{1-\bar{\eta}(K_g)}{K_g}. \tag{A.4}$$

Moreover, using $\Sigma_2(K_g,\bar{\eta}(K_g)) = 0$ and $\Sigma_2(K_g,\eta) = B\Delta(K_g,\eta) + F$,

$$\Delta(K_g,\bar{\eta}(K_g)) = -\frac{F}{B}. \tag{A.5}$$

Using (A.4) and (A.5), (A.3) becomes

$$\Gamma'(K_g) = -Z\int_{\bar{\eta}(K_g)}^{1} (1-\eta)\,dH(\eta) + \frac{F}{B}\frac{1-\bar{\eta}(K_g)}{K_g}h(\bar{\eta}(K_g)). \tag{A.6}$$

The first term is negative, while the second term is positive.

Now evaluate $\Gamma'(K_g)$ at any zero $\hat{K}_g$ of Γ. The zero condition is

$$0 = \Gamma(\hat{K}_g) = \int_{\bar{\eta}(\hat{K}_g)}^{1} \Big[(Z-z)k - Z(1-\eta)\hat{K}_g\Big]\,dH(\eta).$$

Equivalently,

$$\int_{\bar{\eta}(\hat{K}_g)}^{1} (1-\eta)\, dH(\eta) = \frac{(Z-z)k\left[1 - H(\bar{\eta}(\hat{K}_g))\right]}{Z\hat{K}_g}.$$

Substituting this expression into equation (A.6) yields

$$\begin{aligned}
\Gamma'(\hat{K}_g) &= -\frac{(Z-z)k\left[1 - H(\bar{\eta}(\hat{K}_g))\right]}{\hat{K}_g} + \frac{F}{B}\frac{1-\bar{\eta}(\hat{K}_g)}{\hat{K}_g} h(\bar{\eta}(\hat{K}_g)) \\
&= \frac{1}{\hat{K}_g}\left\{\frac{F}{B}(1-\bar{\eta}(\hat{K}_g))h(\bar{\eta}(\hat{K}_g)) - (Z-z)k\left[1 - H(\bar{\eta}(\hat{K}_g))\right]\right\}.
\end{aligned}$$

The sign of $\Gamma'(\hat{K}_g)$ is the sign of the expression in braces. Therefore, $\Gamma'(\hat{K}_g) < 0$ if and only if

$$\frac{B(Z-z)k}{F} > (1-\bar{\eta}(\hat{K}_g))R_H(\bar{\eta}(\hat{K}_g)),$$

where the hazard rate is $R_H(\eta) \equiv h(\eta)/(1-H(\eta))$. Under this condition, every zero of Γ is a downcrossing. Suppose that Γ takes both positive and negative values over the admissible range of K_g. By continuity, Γ has at least one zero. Since it has at most one zero, there exists a unique $\hat{K}_g$ satisfying $\Gamma(\hat{K}_g) = 0$. $\mathbb{Y}$ is uniquely maximized at $\hat{K}_g$.

A.7 Total Derivative with an Endogenous Matching Rate

Proposition 2 holds the matching probability μ fixed to focus on the mode-choice trade-off. This appendix shows how the congestion term enters when the matching probability responds to the mass of searching foreign projects.

Let the matching function μ respond to n. The total derivative of $\mathbb{Y}$ with respect to K_g is

$$\left.\frac{d\mathbb{Y}}{dK_g}\right|_{N,B} = \left.\frac{d\mathbb{Y}}{dK_g}\right|_{\mu,N,B} + \frac{\partial \mathbb{Y}}{\partial \mu}\frac{\partial \mu}{\partial n}\frac{\partial n}{\partial K_g}.$$

The second term, $\frac{\partial \mathbb{Y}}{\partial \mu}\frac{\partial \mu}{\partial n}\frac{\partial n}{\partial K_g}$, is negative because $\partial \mathbb{Y}/\partial \mu > 0$ (a higher matching probability raises the effective input aggregate), $\partial \mu/\partial n < 0$ (search congestion), and $\partial n/\partial K_g = Mg(K_g) > 0$. This term is the congestion externality. Individual foreign projects take μ as given when deciding whether to search, but a planner internalizes the fact that additional search reduces the matching probability for other searchers. Therefore, relative to the fixed-μ benchmark, the planner would choose a narrower search region. This force pushes the decentralized equilibrium toward excessive search and too much M&A.

Appendix B Data Appendix

B.1 Project-Level FDI Data

B.1.1 fDi Markets Database

The Financial Times' fDi Markets database (https://www.fdimarkets.com/) records publicly announced cross-border greenfield investment projects. Financial Times compiles and updates the database using information from its newswires, internal sources, and other media reports.

The database contains 232,000 projects made by firms in 176 source countries and 202 destination countries from 2003 to 2018. For each project, the database reports parent-company information and project-level characteristics, including the parent company name, source and destination locations, project date, industry sector, sub-sector, capital investment, job creation, and project type. In the raw data, source and destination locations are recorded at the city level. My analysis uses the source country, destination country, and industry classification of the project. Around 80% of the reported capital-investment values and job-creation figures are estimated rather than directly observed. For this reason, I focus on the number of investment projects in the paper.

I make two adjustments to the raw data. First, if a company makes more than one investment in several cities within the same destination country on the same project date, the database records them as separate observations. I aggregate such observations to avoid counting what appears to be a single project multiple times. Second, I assign a unique NAICS 2007 code to each project sub-sector based primarily on the concordance provided by the Financial Times, with a small number of manual corrections to address apparent miscategorizations. This step harmonizes the greenfield data with the industry classifications used in SDC Platinum and Compustat. This project-level information is later aggregated to the parent-firm–destination-country–affiliate-industry level used in the estimation sample.

B.1.2 SDC Platinum

The cross-border M&A data come from SDC Platinum. I restrict attention to completed cross-border deals involving US ultimate acquirers. SDC reports both the announcement date (*date announced*) and the completion date (*date effective*). Because the fDi Markets database reports the date when a greenfield project is initiated rather than completed, I use the announcement date in SDC Platinum so that the timing of M&A deals is as comparable as possible to the timing of greenfield investments.

If an acquirer purchases the same target multiple times over the sample period, I aggregate these transactions at the acquirer-target level by summing the ownership shares across deals and keep the year of the first acquisition of that target by the acquirer. I then retain deals in which the acquirer obtains more than 10% ownership of the target.

I exclude deals involving investment funds and government-related entities. Specifically, I drop deals in which either the target or the acquirer is classified in the following three-digit SDC Platinum NAICS codes or special categories: 523 (Securities, Commodity Contracts, and Other Financial Investments and Related Activities), 525 (Funds, Trusts, and Other Financial Vehicles), BCC (Blank Check Company, Sovereign Wealth Fund, Infrastructure Fund, Hedge Fund, and Financial Sponsor), AAA (government-related and public-sector entities), and 92 (Public Administration).

Finally, SDC Platinum contains a small number of special industry codes that do not correspond directly to NAICS 2007. To harmonize the M&A data with Compustat and the greenfield data, I replace these codes as follows: BBBBBA (Internet Service Providers, such as Comcast Corporation) is mapped to NAICS 517911, and BBBBBB (Web Search Portals, such as Alphabet Inc.) is mapped to NAICS 518210.

B.2 Compustat North America and Firm Characteristics

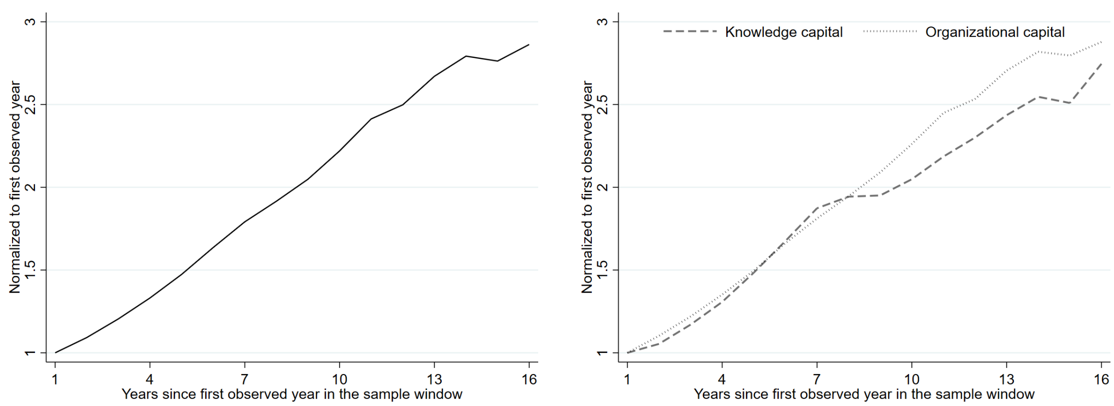
I obtain parent-firm financial information from Compustat North America—Annual Updates. The data cover fiscal years from 1980 to 2018. When the fiscal year is missing, I use the *data date*. I restrict attention to US firms by excluding firms that report financial statements in Canadian dollars and firms headquartered outside the United States. Following Peters and Taylor (2017), I also exclude firms with negative sales. To construct intangible capital using a sufficiently long expenditure history, I further restrict attention to firms that are observed in Compustat for at least six years. All financial variables used in the analysis are deflated using the Consumer Price Index for All Urban Consumers (CPI-U), normalized to 100 in 2002.

Because organizational capital is constructed from SG&A expenditures, observations with missing SG&A do not permit the construction of the baseline intangible-capital measure. I therefore exclude these observations rather than imputing missing SG&A as zero. For R&D expenditures, Compustat reports both zero and missing values. In the baseline analysis, I do not treat missing values as zero, since missing R&D may reflect reporting conventions rather than the absence of innovative activity.

Following Peters and Taylor (2017) and Ewens et al. (2025), I use reported R&D expenditures (*xrd*) as the input to knowledge capital and construct an adjusted SG&A measure for the organizational-capital component. Specifically, I define the R&D-related component as the sum of reported R&D expenditures (*xrd*) and in-process R&D expense (*rdip*), setting *rdip* to zero when missing, and subtract it from SG&A (*xsga*) unless R&D appears already to be included in cost of goods sold.

To harmonize the Compustat data with the industry classifications used in SDC Platinum, I convert Compustat industry codes from NAICS 2017 to NAICS 2007 using the historical NAICS variable (*naicsh*). When the historical code is missing, I assign the corresponding NAICS 2007 code manually. A small number of firms in Compustat are classified under the code 9999 (unclassified establishment), which does not correspond to a valid NAICS industry. For these firms, I assign an industry code using the acquirer's NAICS code in SDC Platinum when the firm undertakes M&A investment; otherwise, I refer to the firm's SEC filings.

Figure B.3: Evolution of Measured Intangible Capital within the Sample Window



(a) Total Intangible Capital (b) Components of Intangible Capital

Notes: Panel (a) plots the average evolution of total measured intangible capital for firms in the sample, using only Compustat observations from 2002 to 2017. Panel (b) plots the corresponding average evolution of knowledge capital and organizational capital. Each series is normalized to one in the firm's first observed year within this sample window.

In addition to intangible capital, I construct value added per worker following İmrohoroğlu and Tüzel (2014). I first compute operating income before depreciation (oibdp) as sales minus cost of goods sold and SG&A expenses. Labor expense is proxied by multiplying the national average wage index from the Social Security Administration by the number of employees. I then define value added as operating income before depreciation (oibdp) plus labor expense and divide it by the number of employees to obtain value added per worker.

Because the stock measures are constructed recursively from past R&D and SG&A expenditures, they vary over time within firms. Appendix Figure B.3 shows their evolution in real terms over the 2002–2017 Compustat window used in the empirical analysis. The series are normalized to one in the firm's first observed year within this sample window. All three measures increase gradually over firms' observed years in the sample period, consistent with the cumulative nature of the stock construction. This upward-sloping pattern is broadly consistent with prior evidence that intangible intensity has risen over time (Peters and Taylor, 2017; Ewens et al., 2025).

B.3 Additional Discussion of the Intangible-Capital Measure

This section provides the definition of intangible capital, the background behind the stock-construction methodology, and three validation concerns.

B.3.1 Definition of intangible capital

Intangible capital consists of non-physical assets based on knowledge or information that generate future economic benefits. In practice, it includes investments that develop specific products or production processes, build organizational capabilities, or strengthen product platforms that allow firms to compete in particular markets (Hulten, 2010). Legally protected forms of intellectual property include patents, trademarks, and copyrights, although intangible capital also includes less formal assets such as data, brands, organizational know-how, and firm-specific capabilities.

B.3.2 Intangible investment and stock construction

Intangible capital is developed within the firm and, unlike physical capital, generally does not appear on the balance sheet. Measuring intangible capital and the associated investment is therefore difficult (Haskel and Westlake, 2017). The literature has developed two broad approaches to measuring intangible capital. One approach uses national accounts to construct aggregate or industry-level measures, and I use this approach for the host-country–affiliate-industry intangible variable. The other uses firm-level accounting data, and I use this approach for the Compustat-based parent-firm intangible stock. In both approaches, intangible-capital stocks are constructed using the perpetual-inventory method and therefore require assumptions about depreciation and the mapping from observed expenditures to intangible investment.

National-account measure

Corrado, Hulten, and Sichel (2005; 2009) develop a methodology for measuring intangible capital using national accounts. They classify intangible capital into three broad categories: computerized information, innovative property, and economic competencies. The first category is measured using aggregate spending on computerized information from the US National Income and Product Accounts. The second includes scientific and non-scientific R&D, measured using the National Science Foundation Industrial R&D series and the Census Bureau's Services Annual Survey (SAS). The third category, economic competencies, includes marketing, brand equity, and organizational and human competencies. This category is measured using advertising expenditures (from the Coen report), training costs (from the Bureau of Labor Statistics surveys), and human and structural resources proxied by revenues of the management consulting industry and the compensation of workers in executive occupations (from SAS). This categorization of intangible capital has also been adopted by the OECD and the European Union, and corresponding data for some countries are available in the EUKLEMS & INTANProd database (Corrado et al., 2022).

Firm-level measure

Building on Corrado et al. (2005; 2009), a recent literature measures firm-level intangible capital using Compustat data. These studies treat R&D and part of SG&A as inputs into the accumulation of intangible assets (Eisfeldt and Papanikolaou, 2013; Belo et al., 2014; Eisfeldt and Papanikolaou, 2014; Gourio and Rudanko, 2014; Peters and Taylor, 2017; Falato et al., 2022; Ewens et al., 2025). R&D expenditures are interpreted as investment in technology, patents, and software, while part of SG&A is interpreted as investment in brand capital, employee training, organizational know-how, and distribution or customer relationships. The stock accumulated from R&D is referred to as knowledge capital, while the stock accumulated from SG&A is referred to as organizational capital.

B.3.3 Validation concerns

I focus on the firm-level intangible-capital measure because, unlike the host-country measure, it is constructed directly from Compustat data for this paper and is the main variable of interest in the empirical analysis.

1: How well do R&D and SG&A capture intangible investment?

A first concern is how accurately the observed expenditures capture the underlying investment that contributes to intangible-capital accumulation. R&D expenditures appear more directly connected to knowledge capital than SG&A appears to organizational capital. There is no direct operational measure of organizational capital, however, and SG&A remains the most widely used proxy for business investment in firm-specific human and marketing resources. SG&A includes many expenditures that plausibly contribute to organizational capital, such as software, employee training, brand enhancement and advertising, payments to systems and strategy consultants, and the cost of setting up and maintaining supply and distribution channels (Lev and Radhakrishnan, 2005).

At the same time, not all SG&A should be interpreted as investment. Earlier work such as Hulten and Hao (2008), Eisfeldt and Papanikolaou (2014), and Peters and Taylor (2017) treats 30% of SG&A as investment in organizational capital and the remaining 70% as operating cost. Ewens et al. (2025) instead estimate industry-specific SG&A shares using information from market prices of intangible capital at firm exits such as acquisitions.[56] Because there is no single accepted benchmark for this fraction, I report results using both the Ewens et al. (2025) methodology and the Peters and Taylor (2017) methodology. The coefficient on Peters and Taylor's organizational capital in column 4 of Table C.2 is numerically identical to the coefficient on organizational capital constructed using Ewens et al. (2025), at the reported level of precision.

I also provide a validation exercise using market valuations. Specifically, I examine whether measured organizational capital is associated with firms' market-to-book ratios. Table B.1 reports firm-year regressions using two outcome variables: the log market-to-book asset ratio and the log equity market-to-book ratio. Across specifications, organizational capital intensity is positively and statistically significantly related to market-to-book ratios. The positive relationship holds when organizational capital is scaled by sales, and it also holds when organizational capital is scaled by book assets while controlling for firm size. The relationship is robust to including fiscal-year fixed effects and industry fixed effects. This pattern supports the interpretation that the SG&A-based measure captures intangible assets valued by financial markets but not fully reflected in firms' book values. I interpret this exercise as a validation check rather than as evidence of a causal effect.

The literature provides additional validation evidence. Organizational capital and total intangible capital are strongly correlated with real indicators of firm performance and quality. Firms with higher organizational capital are more likely to have better brand quality (Ewens et al., 2025), higher scores on the management-quality measure of Bloom and Van Reenen (2007) (Eisfeldt and Papanikolaou, 2013), and disclosures in 10-K filings relating to the loss of key personnel (Eisfeldt and Papanikolaou, 2013; Ewens et al., 2025). Total intangible capital is also strongly correlated with patent valuations and the count of new trademarks (Ewens et al., 2025). Overall, while SG&A and organizational capital are not perfect measures of firms' organizational quality, they are informative and widely used indicators in economic research.

[56] Ewens et al. (2025) report substantial cross-industry variation in the fraction of SG&A attributed to organizational-capital formation, with the highest share in health-related industries (51%) and the lowest in consumer industries (20%).

Table B.1: Organizational Capital and Market Valuation

	Market-to-book assets			Equity market-to-book		
	(1)	(2)	(3)	(4)	(5)	(6)
OrgCap/Sales	0.173***			0.056***		
	(0.009)			(0.010)		
OrgCap/Assets		0.175***	0.171***		0.153***	0.086***
		(0.005)	(0.011)		(0.006)	(0.013)
Sales		-0.041***	-0.034***		0.035***	0.039***
		(0.004)	(0.004)		(0.005)	(0.005)
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry FEs	No	No	Yes	No	No	Yes
Observations	68,014	68,014	67,992	62,347	62,347	62,323

Notes: The unit of observation is a firm-year. The dependent variable in columns 1–3 is the log market-to-book asset ratio, defined as $\log[(AT - CEQ + MVE)/AT]$, where AT is book assets, CEQ is book common equity, and MVE is the market value of equity. The dependent variable in columns 4–6 is the log equity market-to-book ratio, defined as $\log(MVE/CEQ)$. Organizational capital is constructed using the perpetual-inventory method described in Section 3. All variables are in logs. Standard errors, reported in parentheses, are clustered by parent firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

2: Why interpret these expenditures as capital?

A second concern is whether these expenditures should be treated as capital at all. The key question is why expenditures on intangibles should be capitalized rather than treated purely as current expenses (Corrado and Hulten, 2010). R&D, brand equity, and organizational development are aimed at future production and profit rather than only current production and profit (Corrado et al. 2005; 2009). A related interpretation is that the measured intangible stock can be viewed as the replacement cost of the intangible assets that firms have accumulated through these expenditures (Peters and Taylor, 2017). For the empirical question in this paper, the relevant object is the stock of intangible assets available to the firm in the year before it undertakes FDI. This is why I treat the stock measure as the preferred empirical counterpart to the model, although it can also be interpreted as a replacement-cost measure of the intangible assets accumulated through those expenditures or, more broadly, as an indicator of firms' intangible-investment intensity.

3: Choice of depreciation rates

A final concern is the choice of depreciation rates used in the perpetual-inventory method.[57]

[57] The SG&A capitalization share has limited effects on the organizational-capital regressions because it primarily rescales the measure, whereas alternative depreciation rates change the dynamics of the constructed

In the main analysis, I use the most recent parameter estimates from Ewens et al. (2025): industry-specific SG&A shares, industry-specific depreciation rates for knowledge capital, and a constant 20% depreciation rate for organizational capital. As a robustness check, I also construct the stock measures using the parameters in Peters and Taylor (2017), which use a constant 30% SG&A share and a constant 15% depreciation rate for knowledge capital (Table C.2). I further experiment with alternative organizational-capital depreciation rates of 10% and 30% and knowledge-capital depreciation rates of 10% and 20%. All coefficients of intangible capital remain negative and statistically significant (Table C.3).

stock.

B.4 Sample Construction Details

B.4.1 Subsequent Investments

Table B.2 shows the relationship between the entry mode in the first FDI and that in the subsequent FDIs made in the same country and industry. In parent-firm–destination-country–affiliate-industry cells, there are 9,084 cells whose first observed mode is GFDI and 6,460 cells whose first observed mode is M&A. Among cells whose first observed mode is GFDI, 96% are never followed by M&A; among cells whose first observed mode is M&A, 95% are never followed by GFDI.

Table B.2: Persistence of Entry Mode within Destination-Country–Industry Cells

	Subsequent FDI records in the same cell				
First observed mode	Same mode only	Other mode only	Both modes	None	Total
GFDI	**1,859**	194	158	6,873	9,084
M&A	**671**	227	96	5,466	6,460

Notes: Each observation is a parent-firm–destination-country–affiliate-industry cell. The row variable records the first observed entry mode in that cell. The columns classify subsequent foreign investment episodes in the same cell according to whether they use the same mode only, the other mode only, both modes, or no additional investment.

B.4.2 Construction of the Estimation Sample

Table B.3 documents the construction of the estimation sample. The sample is defined by observed FDI projects in fDi Markets and SDC Platinum, not by the universe of Compustat firms. Panel A starts from raw US greenfield projects in fDi Markets, excludes expansions and joint ventures, and then reports the subset of projects whose parent firms can be matched to Compustat. Panel B reports the corresponding construction for cross-border M&A deals in SDC Platinum. Panel C combines the matched GFDI and M&A projects and then aggregates them to the parent-firm–destination-country–affiliate-industry level used in the regressions. The final estimation sample contains 15,544 cell-level observations corresponding to 2,471 Compustat-matched US parent firms. For reference, before matching to the FDI-project data, the Compustat sample used to construct parent-firm intangible capital contains 7,404 unique firms. Compustat firms not matched to any observed foreign-investment transaction in fDi Markets or SDC Platinum during the sample period are outside the estimation sample by construction.

Table B.3: Construction of the Estimation Sample

	Number of FDI projects	Number of US parent firms
Panel A: fDi Markets		
Raw US GFDI projects	43,071	12,373
After excluding expansions/JVs	38,555	12,003
Matched to Compustat	14,851	1,771
Panel B: SDC Platinum		
Raw cross-border M&A deals	25,898	8,483
Matched to Compustat	8,209	1,858
Panel C: Final estimation sample		
Combined Matched FDI projects	23,060	2,479
Final Sample at the cell level	15,544	2,471

Notes: The estimation sample is defined by observed FDI projects in fDi Markets and SDC Platinum. Compustat is used only to append parent-firm financial variables for the subset of investing firms that can be matched. Parent-firm counts across fDi Markets and SDC Platinum are not additive because some firms appear in both data sources.

B.5 Descriptive Patterns in the Matched FDI Sample

B.5.1 Descriptive Decomposition of Entry-Mode Variation

Prior work reaches different conclusions about whether FDI mode choice is driven primarily by firm, country, or industry characteristics. To assess the relative importance of these dimensions in my data, I follow Coşar and Demir (2018) and decompose the variation in entry mode across parent-firm–destination-country–affiliate-industry cells.

Let $MA_{i,h,a,t}$ be an indicator equal to one if parent firm i uses M&A for its first observed entry in destination country h and affiliate industry a in year t, and zero if it uses GFDI. Appendix Table B.4 reports partial-determination measures comparing the explanatory power of parent-firm, destination-country, and affiliate-industry components.[58] To construct the first column, I compare the explanatory power of firm fixed effects with the best-performing specification that does *not* include firm-level effects—namely, the model with country-industry fixed effects. Specifically, I first regress $MA_{i,h,a,t}$ on firm fixed effects and country-industry pair fixed effects (i.e., individual fixed effects and pair fixed effects of the remaining factors) and obtain the adjusted $R^2_{i,ha}$. I next regress $MA_{i,h,a,t}$ only on country-industry pair fixed effects (without firm fixed effects) and keep the adjusted R^2_{ha}. I then obtain the share of variation that firm fixed effects can explain by computing $(R^2_{i,hj} - R^2_{ha})/(1 - R^2_{ha})$. The other columns are constructed analogously.

The results show that parent-firm heterogeneity explains a larger share of the variation in cell-level entry mode than destination-country or industry factors. I view this pattern as motivating the paper's focus on parent-firm characteristics in the empirical analysis, while maintaining the parent-firm–destination-country–affiliate-industry cell as the relevant unit of observation.

Table B.4: Sources of Variation in Firm FDI Mode Decisions

Single fixed effects			Pairwise fixed effects		
Firm$_i$	Country$_h$	Industry$_a$	Firm-country$_{ih}$	Firm-ind$_{ia}$	Country-ind$_{ha}$
0.204	0.135	0.147	0.363	0.410	0.270

Notes: This table reports partial-determination measures following Coşar and Demir (2018). The unit of observation is at the parent-firm–destination-country–affiliate-industry level.

[58] I exclude year fixed effects because they explain very little variation. For example, the direct effect of year variation is 0.011.

B.5.2 FDI Destinations

This table shows the five major FDI destinations in terms of the number of FDI investments they received between 2003 and 2018. China and India attract more GFDI, while the UK, Canada, and Germany attract cross-border M&A.

Table B.5: Top Five FDI destinations

	All FDI		Greenfield		M&A	
Rank	Country	Share (%)	Country	Share (%)	Country	Share (%)
1	UK	10.04	China	8.94	UK	16.67
2	Canada	7.78	India	6.54	Canada	14.16
3	China	6.27	UK	5.33	Germany	7.97
4	Germany	5.44	Singapore	3.67	Australia	5.68
5	India	4.91	Germany	3.64	France	5.43

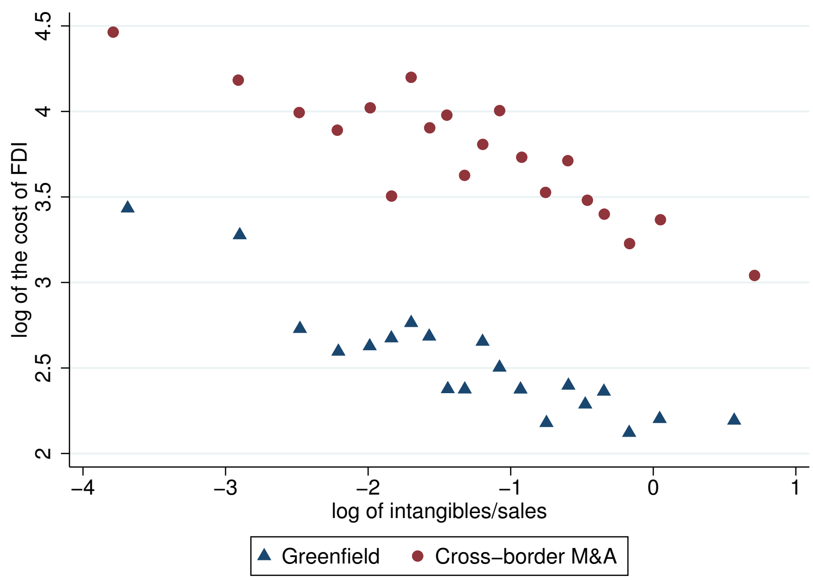
Notes: This table shows the five most common FDI destinations of US multinational firms. I rank the countries in terms of the number of FDI investments they received between 2003 and 2018. The first two columns show the names of host countries and the share of all FDI investments by US-listed firms that went to that location. The following columns show the same information by splitting FDI into greenfield investment and cross-border M&A.

B.5.3 Transaction Size by Parent-Firm Intangible Intensity

The FDI project data contain transaction-size information, but the reported size measure differs across entry modes. For greenfield projects, fDi Markets reports capital-investment values. For cross-border M&A deals, SDC Platinum reports acquisition values. These two variables should therefore not be interpreted as directly comparable fixed costs of foreign entry. I use them only as a descriptive check.

Acquisition values are available for 58% of M&A deals in my data. Capital-investment values are available for almost all greenfield projects, although many are estimated by the Financial Times. The average GFDI project is worth \$61 million, while the average M&A transaction is worth \$258 million. Acquisition values remain larger than greenfield capital expenditures throughout the distribution of parent-firm intangible intensity (Figure B.4). This pattern is consistent with the idea that acquisitions embody assets beyond physical capital.

Figure B.4: Transaction Size by Parent-Firm Intangible Intensity



Notes: The figure plots binned averages of log transaction size against log parent-firm intangible intensity, measured as intangible capital over sales. For greenfield projects, transaction size is the capital-investment value reported in fDi Markets. For cross-border M&A deals, transaction size is the acquisition value reported in SDC Platinum.

Appendix C Additional Figures and Tables

C.1 Prior Foreign Presence and FDI Entry-Mode Results

Since the fDi Markets database begins in 2003, a potential concern is that a parent firm's observed project in my sample may not coincide with its first foreign presence in a destination country. To address this concern, I use the WRDS company subsidiary data to identify prior foreign affiliates before 2003 or before the firm first appears in Compustat. Because these supplemental data identify prior foreign presence only at the country level, and do not report the affiliate industry's classification, I impose this restriction conservatively by excluding all observations in destination countries with prior foreign presence.

Appendix Table C.1 reports the results under this restriction. The coefficient on total intangible capital remains negative and statistically significant across specifications. The organizational-capital and knowledge-capital coefficients are also negative and statistically significant. These results suggest that the main finding is not driven by unobserved foreign presence in destination countries before the start of the sample period or before the parent firm first appears in Compustat.

Table C.1: Prior Foreign Presence and FDI Entry Mode

	Total intangibles			Organizational	Knowledge	Physical
Dep. var: M&A indicator	(1)	(2)	(3)	(4)	(5)	(6)
Intangible capital	-0.021***	-0.047***	-0.052***			
	(0.003)	(0.009)	(0.011)			
Organizational capital				-0.035***		
				(0.009)		
Knowledge capital					-0.029***	
					(0.010)	
Physical capital						-0.001
						(0.009)
Sales		0.018*	0.023**	0.007	0.003	-0.025**
		(0.010)	(0.011)	(0.010)	(0.011)	(0.011)
Value-added per worker		0.018	0.016	0.017	0.025*	0.019
		(0.011)	(0.013)	(0.011)	(0.013)	(0.012)
Country $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry-pair FEs	Yes	Yes	No	Yes	Yes	Yes
Industry-pair $\times$ Year FEs	No	No	Yes	No	No	No
Observations	10,331	9,751	8,056	9,751	5,597	9,607

Notes: The unit of observation is a parent-firm–destination-country–affiliate-industry cell. The dependent variable equals one if the first observed FDI in the cell is an M&A. All parent-firm financial variables are lagged by one year relative to the entry-mode decision. Standard errors, reported in parentheses, are clustered by parent firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Alternative Intangible-Capital Construction

The baseline analysis measures internally generated intangible capital following Ewens et al. (2025). As an additional robustness check, Appendix Table C.2 reconstructs intangible capital using the methodology of Peters and Taylor (2017). Following Peters and Taylor (2017), I treat 30% of SG&A expenditure as investment in organizational capital and use the 15% depreciation rate of knowledge capital.[59]

The results of Appendix Table C.2 preserve the main pattern in Table 2: the coefficient on parent-firm intangible capital remains negative and statistically significant in the baseline specification, in the specification with parent-firm controls, and in the specification with industry-pair–year fixed effects. When the stock is decomposed into organizational capital and knowledge capital, both components enter negatively and are statistically significant.

Table C.2: Alternative Intangible-Capital Construction: Peters and Taylor (2017)

	Total intangibles			Organizational	Knowledge
Dep. var: M&A indicator	(1)	(2)	(3)	(4)	(5)
Intangible capital (P&T)	-0.021***	-0.039***	-0.046***		
	(0.003)	(0.008)	(0.009)		
Organizational capital (P&T)				-0.032***	
				(0.009)	
Knowledge capital (P&T)					-0.018**
					(0.009)
Sales		0.010	0.018*	0.003	-0.009
		(0.009)	(0.010)	(0.009)	(0.010)
Value-added per worker		0.018*	0.016	0.015	0.019
		(0.011)	(0.012)	(0.011)	(0.013)
Country × Year FEs	Yes	Yes	Yes	Yes	Yes
Industry-pair FEs	Yes	Yes	No	Yes	Yes
Industry-pair × Year FEs	No	No	Yes	No	No
Observations	14,653	14,024	11,914	14,024	8,434

Notes: The unit of observation is a parent-firm–destination-country–affiliate-industry cell. The dependent variable equals one if the first observed FDI in the cell is an M&A. All parent-firm financial variables are lagged by one year relative to the entry-mode decision. Standard errors, reported in parentheses, are clustered by parent firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[59] Peters and Taylor (2017) report that their results are virtually unchanged when fixed R&D depreciation rates of 10%, 15%, or 20% are used.

Appendix Table C.3 further examines whether the results are sensitive to the depreciation rates used to construct the intangible-capital stocks. Columns 1–3 vary the depreciation rate for organizational capital, using 10%, 20%, and 30%, respectively, with the 20% rate corresponding to the Peters and Taylor (2017) benchmark. The coefficient on organizational capital is negative and statistically significant across all three specifications. Columns 4–6 vary the depreciation rate for knowledge capital, using 10%, 15%, and 20%, respectively, with the 15% rate corresponding to the Peters and Taylor (2017) benchmark. The coefficient on knowledge capital is also negative across all three specifications and remains statistically significant.

Overall, these findings indicate that the main result does not depend on the particular intangible-capital stock-construction method or depreciation-rate assumption used in the baseline analysis.

Table C.3: Alternative Intangible-Capital Construction: Different Depreciation Rates

	Organizational capital			Knowledge capital		
Dep. var: M&A indicator	(1)	(2)	(3)	(4)	(5)	(6)
	$\delta_{SG\&A} = 0.1$	$\delta^{P\&T}_{SG\&A} = 0.2$	$\delta_{SG\&A} = 0.3$	$\delta_{R\&D} = 0.1$	$\delta^{P\&T}_{R\&D} = 0.15$	$\delta_{R\&D} = 0.2$
Organizational capital	-0.023***	-0.032***	-0.038***			
	(0.008)	(0.009)	(0.010)			
Knowledge capital				-0.015*	-0.018**	-0.020**
				(0.008)	(0.009)	(0.009)
Sales	-0.005	0.003	0.008	0.012	-0.009	0.007
	(0.008)	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)
Value-added per worker	0.015	0.015	0.015	0.019	0.019	0.020
	(0.011)	(0.011)	(0.011)	(0.013)	(0.013)	(0.013)
Country $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry-pair FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,024	14,024	14,024	8,434	8,434	8,434

Notes: The unit of observation is a parent-firm–destination-country–affiliate-industry cell. The dependent variable equals one if the first observed FDI in the cell is an M&A. All parent-firm financial variables are lagged by one year relative to the entry-mode decision. Standard errors, reported in parentheses, are clustered by parent firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Robustness to Alternative Restrictions on Repeated Entry Modes

As discussed in Section 3.3, mode switching within a parent-firm–destination-country–affiliate-industry cell is rare: only 4.3% of cells exhibit both GFDI and M&A over the sample period, and the first observed mode is highly persistent within a market. Nevertheless, I further assess whether the baseline result is sensitive to repeated observed entry or to the use of both FDI modes within the observed sample.

Appendix Table C.4 considers three such restrictions. Column 2 excludes cells that ever exhibit both GFDI and M&A during the sample period. The estimated coefficient on intangible capital remains virtually unchanged relative to the baseline. Column 3 restricts the sample to parent-firm–destination-country–affiliate-industry cells with exactly one observed FDI episode. The coefficient remains negative and statistically significant. Finally, column 4 imposes a more stringent firm-level restriction by keeping only parent firms that use a single FDI mode across all observed destination-country–affiliate-industry markets throughout the sample period. Although this restriction substantially reduces the number of observations, the coefficient remains negative and statistically significant, with a larger absolute magnitude than in the baseline. Taken together, these exercises indicate that the baseline relationship between intangible capital and entry mode is not driven by cells that use both FDI modes, or by parent firms that use both modes across markets.

Table C.4: Robustness to Alternative Treatments of Multiple FDI Projects Within Cells

	Baseline	Excl. cells with both modes	One FDI per cell	Single-mode firms
Dep. var: M&A indicator	(1)	(2)	(3)	(4)
Intangible capital	-0.043***	-0.044***	-0.038***	-0.060***
	(0.008)	(0.008)	(0.009)	(0.021)
Sales	0.013	0.012	0.011	0.034*
	(0.009)	(0.009)	(0.009)	(0.020)
Value-added per worker	0.017	0.018	0.019*	0.034
	(0.011)	(0.011)	(0.011)	(0.024)
Country × Year FEs	Yes	Yes	Yes	Yes
Industry-pair FEs	Yes	Yes	Yes	Yes
Observations	14,024	13,357	10,902	1,913

Notes: The unit of observation is a parent-firm–destination-country–affiliate-industry cell. The dependent variable equals one if the first observed FDI in the cell is an M&A. All parent-firm financial variables are lagged by one year relative to the entry-mode decision. Standard errors, reported in parentheses, are clustered by parent firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

C.4 Firm-Year History Requirement for Intangible-Capital Construction

Because the baseline intangible-capital measure is constructed recursively from past R&D and SG&A expenditures, a natural concern is that firm-years with shorter observed expenditure histories may be measured with more noise. As a robustness check, Appendix Table C.5 imposes a stricter firm-year history requirement, under which a firm-year enters the regression sample only after six years of expenditure history are available for stock construction. The coefficient on total intangible capital remains negative and statistically significant under the stricter firm-year history requirement, with magnitudes close to those in the corresponding specifications in Table 2. These findings indicate that the baseline relationship between parent-firm intangible capital and FDI entry mode is not driven by firms with shorter observed expenditure histories.

Table C.5: Robustness to a Stricter Firm-year History Requirement

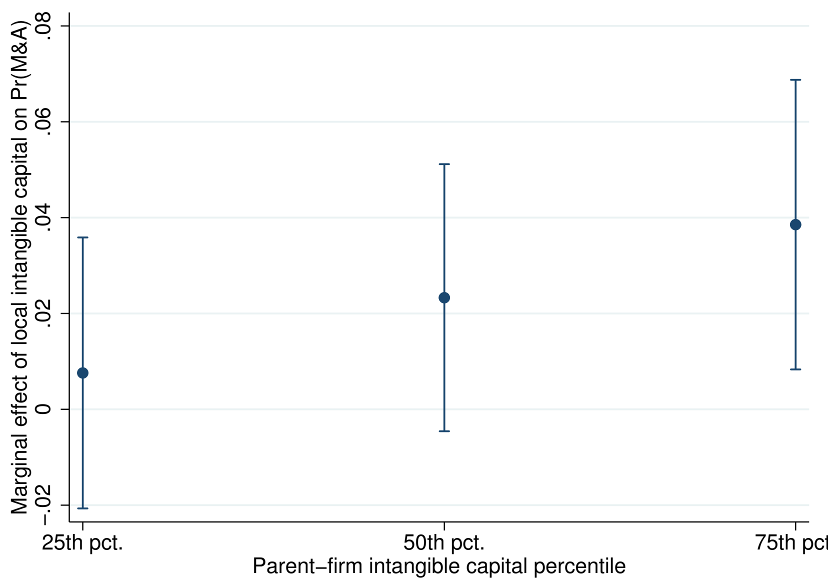
	Total intangibles			Organizational	Knowledge	Physical
Dep. var: M&A indicator	(1)	(2)	(3)	(4)	(5)	(6)
Intangible capital	-0.031***	-0.047***	-0.050***			
	(0.003)	(0.009)	(0.011)			
Organizational capital				-0.037***		
				(0.010)		
Knowledge capital					-0.024**	
					(0.010)	
Physical capital						-0.013
						(0.010)
Sales		0.013	0.016	0.003	-0.008	-0.017
		(0.010)	(0.011)	(0.010)	(0.011)	(0.011)
Value-added per worker		0.017	0.015	0.014	0.019	0.017
		(0.012)	(0.014)	(0.012)	(0.014)	(0.013)
Country × Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Industry-pair FEs	Yes	Yes	No	Yes	Yes	Yes
Industry-pair × Year FEs	No	No	Yes	No	No	No
Observations	12,727	12,486	10,504	12,486	7,459	12,369

Notes: The unit of observation is a parent-firm–destination-country–affiliate-industry cell. The dependent variable equals one if the first observed FDI in the cell is an M&A. All parent-firm financial variables are lagged by one year relative to the entry-mode decision. Standard errors, reported in parentheses, are clustered by parent firm. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Marginal Effect of Local Intangibles by Parent intangibles

Appendix Figure C.1 illustrates the interaction result reported in column 5 of Table 3. Because the regression includes an interaction between local intangible assets and parent-firm intangible capital, the coefficient on local intangible assets alone corresponds to the effect when parent-firm intangible capital equals zero and is therefore not economically meaningful on its own. The figure instead reports the marginal effect of local intangible assets on the probability of choosing M&A, evaluated at the 25th, 50th, and 75th percentiles of parent-firm intangible capital. The point estimates are positive and increasing across these percentiles. The effect is imprecisely estimated at lower percentiles but becomes statistically distinguishable from zero for parent firms with high intangible capital. This pattern indicates complementarity between parent-firm intangible assets and local intangible assets in FDI mode choice: local intangible-rich markets are associated with relatively more M&A, especially for parents with greater intangible intensity.

Figure C.1: Marginal effect of local intangibles by parent intangibles



Notes: The figure reports the marginal effect of local intangible assets on the probability of choosing M&A at the 25th, 50th, and 75th percentiles of parent-firm intangible capital. Error bars denote 95% confidence intervals.

C.6 Host-Country Target Availability and Reduced-Country Samples

Because destination-country–affiliate-industry measures of local firm counts and local intangible capital are available only for subsets of host countries and years, the country coverage differs across the columns of Table 3. Appendix Table C.6 examines whether the estimates are affected by these changes in sample coverage. For each host-market variable, I re-estimate the broad-controls specification on the same reduced-country sample used in the corresponding regression in the main text. Columns 1 and 2 compare the local-firm-count specification with the broad-controls specification on the same 44-country sample. Columns 3 and 4 do the same for local intangible capital on the smaller 20-country sample. The coefficients on parent-firm intangible capital and distance remain negative across these reduced-country samples.

Table C.6: Reduced-Country Sample Comparisons for Host-Market Regressions

	Num of local firms (44 countries)		Local intangibles (20 countries)	
Dep. var: M&A indicator	(1)	(2)	(3)	(4)
Parent intangibles	-0.029*	-0.029**	-0.052***	-0.052***
	(0.015)	(0.014)	(0.014)	(0.012)
Distance	-0.117*	-0.078**	-0.547***	-0.540***
	(0.061)	(0.035)	(0.065)	(0.082)
Common language	0.033	0.040**		
	(0.036)	(0.018)		
Num local firms per capita		0.034***		
		(0.011)		
Local intangibles per capita				0.025*
				(0.014)
GDP per capita	0.271***	0.266***	0.367***	0.310***
	(0.034)	(0.022)	(0.046)	(0.047)
Industry-pair FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	5,180	5,180	4,259	4,259

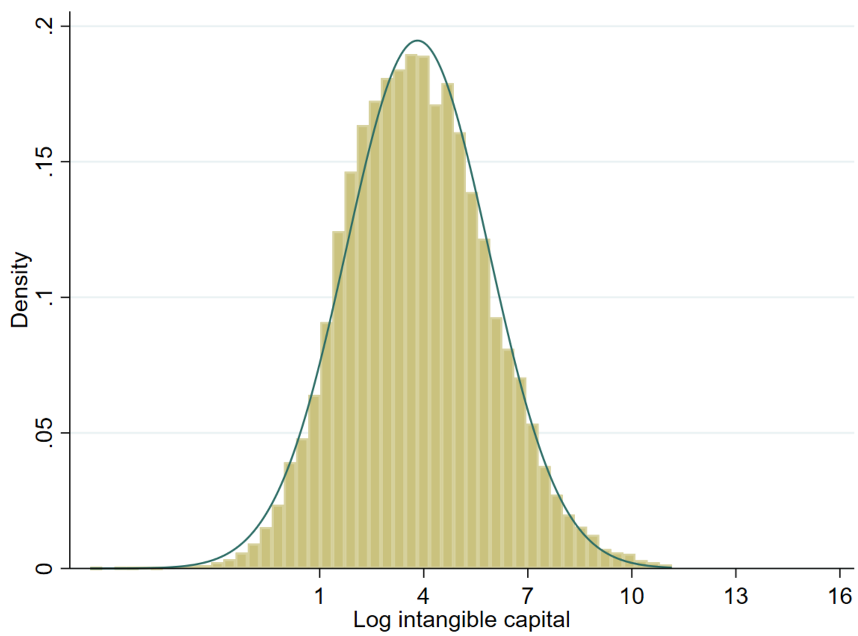
Notes: The dependent variable is the first-observed-entry M&A indicator at the parent-firm–destination-country–affiliate-industry level. All columns include openness, sales, and value-added per worker. Columns 1 and 3 report the broad-controls specification on the reduced samples used in columns 2 and 4, respectively. Standard errors are two-way clustered by parent firm and by destination-country–industry market. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix D Additional Calibration Details

D.1 Distribution of Intangible Capital

This section provides diagnostics for the log-normal specification of the distribution of foreign intangible capital. Figure D.1 plots the distribution of log intangible capital in the potential-parent Compustat firm-year sample together with the fitted normal density. The baseline estimate of the dispersion parameter is $\theta_K = 2.049$. As checks, removing year-specific means of log intangible capital gives $\theta_K = 2.035$, and winsorizing log intangible capital at the 1st and 99th percentiles gives $\theta_K = 2.007$.

Figure D.1: Distribution of log intangible capital



Note: The figure shows the distribution of log intangible capital in the potential-parent Compustat firm-year sample used to estimate the dispersion parameter of $G(K)$. Intangible capital is constructed from deflated R&D and SG&A expenditures. The fitted normal density is overlaid.

D.2 Details of the Calibration Section

D.2.1 Data

US and Destination Productivity. Labor productivity per hour worked reports output per hour worked in constant international dollars and is adjusted for inflation and cross-country differences in purchasing power. Since the model assumes that local project productivity does not exceed the productivity of foreign projects (e.g., $Z > z$), I truncate the destination productivity z above at one. This adjustment mainly affects a small set of resource-intensive or high-income destinations whose aggregate GDP per hour worked exceeds the US level. For example, destinations with productivity more than 10 percent above the US account for 3.6 percent of projects.

Labor Endowment per Potential Foreign Project. The scaling factor ν_M is the ratio of the number of firms in the Compustat sample to the number of those firms observed undertaking FDI.

D.2.2 Numerical Implementation

I impose the interior cutoff ordering associated with Proposition 1: $0 < K_m < \tilde{K} < K_\eta < K_g$. This restriction ensures that the interval in which the realized integration-quality draw can affect the post-match mode choice is nonempty. I impose the ordering by parameterizing

$$K_m = s\tilde{K} \quad \text{where} \quad s \in (0,1), \qquad \text{and} \qquad K_g = K_\eta + \exp(y).$$

The remaining inequality $\tilde{K} < K_\eta$ follows from $Z > z$ and $k > 0$ since $K_\eta = \tilde{K} + (Z - z)k/Z$.

D.3 Decomposing the North–South Difference

To trace the source of the difference in policy implications between the advanced and non-advanced calibrations, I start from the non-advanced-economy calibration and cumulatively replace its parameters with their advanced-economy values. At each step, I recompute the equilibrium and the inefficiency wedge.

Table D.7: Decomposition of the Inefficiency Wedge

	$\psi/[\mu(1-\chi)]$	$F[1-H(\bar{\eta})]$	Wedge, $B\Gamma$	μ	$\Pr(M\&A \mid FDI)$
Non-advanced baseline	0.487	0.361	0.126	0.438	0.148
$+\psi$, z	0.083	0.284	−0.202	0.208	0.324
$+F$	0.052	0.058	−0.006	0.332	0.154
$+k$	0.051	0.058	−0.006	0.337	0.186
$+\xi_1$	0.060	0.058	0.003	0.285	0.194
$+\nu_N$	0.018	0.037	−0.019	0.942	0.923
$+L/M$	0.044	0.057	−0.012	0.387	0.370
$+c$ (full advanced)	0.035	0.054	−0.019	0.517	0.545

Notes: Starting from the non-advanced-economy calibration, each row cumulatively replaces the indicated parameters with their advanced-economy values and recomputes the equilibrium. The first two columns report the two components of the wedge in equation (27); the search-cost component and the fixed-cost-saving component. The parameters ψ and z are replaced jointly in the first step because ψ cannot be replaced in isolation while preserving the ordered equilibrium.

The results are in Table D.7. The decomposition shows that the difference in the inefficiency wedge between the non-advanced- and advanced-economy calibrations is driven mainly by the higher search cost ψ. In the first step, ψ is replaced jointly with local productivity z in order to preserve the ordering of four thresholds $\{K_m, \tilde{K}, K_\eta, K_g\}$. Since z does not enter the search-cost component except through the endogenous matching probability μ, the sharp decline in this component is primarily attributable to the higher search cost.

The wedge is positive in the non-advanced baseline, $B\Gamma = 0.126$, implying too little M&A. Once ψ and z take their advanced-economy values, the search-cost component $\psi/[\mu(1-\chi)]$ falls from 0.487 to 0.083, and the wedge turns negative. Replacing F in the next step reduces the fixed-cost-saving component from 0.284 to 0.058, bringing the wedge close to zero. The remaining parameter replacements move the wedge only modestly. The decomposition confirms the analytical reading of equation (27): the inefficiency in the non-advanced economy is therefore driven by the high search cost, partly offset by the high fixed cost, whereas the advanced-economy calibration lies close to the efficient margin because the two components nearly cancel.

D.4 Introducing Policy Instruments in the Model

This appendix presents the equations affected by an M&A subsidy s_m and a GFDI tax τ_g in the model. With the policy in place, the operating profits in equation (4) become

$$\pi_i^g = BZK_i - F - \tau_g, \qquad \pi_i^m = BZ(k + \eta_i K_i) - V + s_m, \qquad \pi_j = Bzk.$$

The tax raises the cost of GFDI entry from F to $F + \tau_g$, the subsidy adds to the acquirer's payoff from completed M&A, and the standalone local profit is unchanged.

Because the subsidy s_m is paid to the acquirer outside the acquisition negotiation, it is not part of the merger surplus over which the foreign acquirer and the local target bargain. The total merger surplus (8) is modified only by the GFDI tax in the outside option:

$$\Sigma^\tau(K_i, \eta_i) = \begin{cases} \Sigma_1(K_i, \eta_i) \equiv B[(Z - z)k + Z\eta_i K_i] & \text{for } K_i \in (0, \tilde{K}), \\ \Sigma_2^\tau(K_i, \eta_i) \equiv B[(Z - z)k - Z(1 - \eta_i)K_i] + F + \tau_g & \text{for } K_i \in [\tilde{K}, \infty), \end{cases}$$

The GFDI zero profit threshold $\tilde{K}$, the integration threshold K_η, and the completion cutoff $\bar{\eta}(K)$ depend on the GFDI tax:

$$\tilde{K} = \frac{F + \tau_g}{BZ}, \qquad K_\eta = \tilde{K} + \frac{Z - z}{Z}k, \qquad \bar{\eta}(K) = \max\left\{0,\ 1 - \frac{1}{ZK}\left[(Z - z)k + \frac{F + \tau_g}{B}\right]\right\}.$$

The completion threshold $\bar{\eta}$ does not depend on s_m. A matched project completes the merger only if the local target is willing to sell, that is only if $V = \pi_j + \chi\Sigma_2^\tau \geq \pi_j$, or equivalently $\Sigma_2^\tau(K, \eta) \geq 0$. Since the subsidy is paid solely to the acquirer and is not shared with the target in the bargaining problem, it cannot induce the target to accept a negative-surplus merger. The set of matches that complete as M&A is therefore still characterized by $\eta \geq \bar{\eta}(K)$ with $\bar{\eta}$ defined by $\Sigma_2^\tau = 0$, unaffected by s_m. The subsidy changes the acquirer's incentive to enter the merger market, but not which matched projects are completed as M&A.